Abstract Algebra Notes

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1. Groups

1.1. Group Basics

Definition (group): A *group* is a set G with an operation $\times : G \times G \to G$ which satisfies the following properties:

a) There exists an element $e \in G$ such that $a \times e = e \times a = a$ for all $a \in G$ (if the operation is represented multiplicatively, then often we write it as 1_G, or just 1).
b) For every $a \in G$, there exists $a^{-1} \in G$ such that $a \times a^{-1} = a^{-1} \times a = e$.
c) The operation is associative, i.e. for every $a, b, c \in G$, we have $a \times (b \times c) = (a \times b) \times c$.

From here on out, the operation $\times$ will take a number of forms, such as $+$ or $\cdot$, or it won't make an appearance at all and implicit multiplication (such as ab) will represent the operation.

Example *(standard examples/nonexamples)*:

- Consider $(\mathbb{Z}, +)$. The identity is 0, and the inverse of n is $-n$.
- Consider $(\mathbb{Z}^\times, \cdot)$, where $\mathbb{Z}^\times = \mathbb{Z} \backslash \{0\}$. The identity is 1, but not every element has an inverse. For example, $2 \cdot n \neq 1$ for any integer n.
- Consider $(\mathbb{C}^\times, \cdot)$. The identity is 1, and the inverse of z is z^{-1}.
- Consider $(\mathrm{SL}_n(\mathbb{C}), \times)$, where $\mathrm{SL}_n(\mathbb{C})$ is the set of $n \times n$ matrices with elements in $\mathbb{C}$ that have determinant 1. The identity is I_n, and since every element has nonzero determinant, an inverse exists and is given by the inverse matrix.
- Consider $\{1, 2, ..., n\}$ and the set of functions T on it that are bijective. Then S_n denotes the group of all of these functions, and is called the *symmetric group*. The identity permuation is just $f(x) = x$, and the inverse of an element is given by the inverse function.
- The integers mod n form a group under addition, and is what's known as *cyclic*, since a single element (say 1) generates the entire group by repeatedely applying the operation to the element. A group with size n where a single element generates it is often called a *cyclic group*, and is denoted C_n.
- The group G with one element 1 is called the *trivial group*.

Definition (abelian): Suppose $(G, \cdot)$ is a group. If for every $a, b \in G$ we have $ab = ba$, then the group is called *abelian*.

Taking $\mathrm{SO}(n)$ as an example, we see that not every group is abelian.

Proposition (identity and inverse are unique): Let $(G, \cdot)$ be a group. Then the identity is unique. Furthermore, every element has a unique inverse.

Proof: Suppose e_1, e_2 are identities. Then $e_2 = e_1 e_2 = e_1$. Now suppose x, y are inverse of a. Then $x = xe = x(ay) = (xa)y = ey = y$. ■

Definition (left/right inverse): Suppose $(G, \cdot)$ is a group. An element a has a *left inverse* if there exists $\ell \in G$ such that $\ell a = e$. An element a has a *right inverse* if there exists $r \in G$ such that $ar = e$.

Proposition (left and right inverse imply invertible): Suppose $(G, \cdot)$ is a group, and $a \in G$ has left inverse ℓ and right inverse r. Then $\ell = r$ and a is invertible with inverse ℓ.

Proof: We have

$$\ell = \ell(ar) = (\ell a)r = r.$$

Then we have

$$\ell a = e = ar = a\ell.$$

■

Proposition (cancellation law): Suppose $(G, \cdot)$ is a group. If $a, b, c \in G$ such that $ab = ac$ or $ba = ca$.

Proof: Multiply the first equation by a^{-1} on the left, and second equation by a^{-1} on the right.■

Definition (product group): Let $(G, \star)$ and $(H, *)$ be groups. Then the *product group* $(G \times H, \cdot)$ has operation $\cdot$ defined by

$$(g_1, h_1) \cdot (g_1, h_2) = (g_1 \star g_2, h_1 * h_2) \in G \times H.$$

Definition (subgroup): Suppose $(G, \star)$ is a group. Then a subset H of G is a *subgroup of* G, denote $H \leq G$, if when inheriting the operation $\star$, H is a group. If $H \neq G$, then H is a *proper subgroup of* G, denoted $H < G$.

Example *(subgroup generated by element)*: Let x be an element of a group G. Then the subset

$$\langle x \rangle = \{\ldots, x^{-2}, x^{-1}, 1, x^1, x^2, \ldots\}$$

is a subgroup of G. More generally, given a subset S of a group G, we let $\langle S \rangle$ denote the subgroup given by creating any string of the elements of S and their inverses.

Proposition: If $H \leq G$, then H has the same identity as G, and the inverse of an element in H is the same as the elements' inverse in G.

Proof: Let 1_G denote the identity in G, and let 1_H denote the identity in H. Since $H \subseteq G$, we have $h1_G = 1_G h = h$ for all $h \in H$. Thus 1_G is an identity of H. Since the identity of a group is unique, this implies $1_G = 1_H$.

Now suppose $h \in H$ has inverse $h_H^{-1} \in H$ and inverse $h_G^{-1} \in G$. Since $h, h_H^{-1} \in G$, this implies that h_H^{-1} is an inverse of h in G. Since inverses are unique, this implies $h_G^{-1} = h_H^{-1}$. ■

Proposition: The intersection of two subgroups $H_1, H_2 \leq G$ is another subgroup of G.

Proof: Clearly $1 \in H_1 \cap H_2$. If $a, b \in H_1 \cap H_2$, then $a, b \in H_1$ and $a, b \in H_2$, so $ab \in H_1 \cap H_2$ by closure. Similar logic shows that $H_1 \cap H_2$ contains inverses. ■

Definition (order of group and element): The *order* of a group G is the number of elements in it. The *order* of an element $x \in G$ is the smallest $n \in \mathbb{N}$ such that $x^n = 1$, and is denoted ord x. If no such n exists, then the order is ∞.

Proposition: Suppose G is a group. If $g^n = 1$, then ord g divides n.

Proof: Suppose otherwise. Thus $n = a \cdot \text{ord } g + b$, where $b < \text{ord } g$ and $a, b \in \mathbb{Z}_{\geq 0}$. Then we have

$$g^n = g^{a \cdot \text{ ord } g} g^b = g^b.$$

Since ord g is the minimal number such that $g^k = 1$, $g^b \neq 1$, so we have a contradiction. ■

Proposition: Suppose G is a finite group. Then ord g is finite for all $g \in G$.

Proof: If not, then the sequence $g, g^2, g^3, \ldots$ would contain pairwise distinct elements, and thus G would not be finite, contradiction. ■

1.1.1. Isomoprhisms and Homomorphisms

Definition (isomorphism): Let $(G, \times)$ and $(H, \star)$ be groups. A bijection $\varphi : G \to H$ is called an *isomorphism* if

$$\varphi(g_1 \times g_2) = \varphi(g_1) \star \varphi(g_2)$$

for all $g_1, g_2 \in G$. If there exists an isomorphism between G and H, then we say they are *isomorphic*, and denote it $G \cong H$.

Example *(isomorphisms)*:

- The identity map from G to itself is trivially an isomorphism.

- Define the map $\varphi : S_2 \to \mathbb{Z}/2\mathbb{Z}$ by $\varphi(\{1,2\}) = 0$ and $\varphi(\{2,1\}) = 1$. Then it's easy to check that φ is an isomorphism, so $\mathbb{Z}/2\mathbb{Z} \cong S_2$.

Definition (automorphism): If $\varphi : G \to G$ is an isomorphism, then φ is an *automorphism*.

Definition (homomorphism): Let $(G, \times)$ and $(H, \star)$ be groups. A function $\varphi : G \to G$ is called a *homomorphism* if

$$\varphi(g_1 \times g_2) = \varphi(g_1) \star \varphi(g_2)$$

for all $g_1, g_2 \in G$.

Definition (endomorphism): If $\varphi : G \to G$ is a homomorphism, then φ is an *endomorphism*.

Example *(homomorphism)*:
- Every isomorphism is a homomorphism.
- Consider $\varphi : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$ given by $\varphi(n) = n \pmod 3$. Then it's easy to see this is a homomorphism.

Proposition: Let φ be a homomorphism between G and H. Then $\varphi(1_G) = 1_H$ and $\varphi(a)^{-1} = \varphi(a^{-1})$ for all $a \in G$.

Proof: We have

$$\varphi(1_G) = \varphi(1_G \times 1_G) = \varphi(1_G) \star \varphi(1_G) \Rightarrow 1_H = \varphi(1_G).$$

We also have

$$1_H = \varphi(1_G) = \varphi(a \times a^{-1}) = \varphi(a)\varphi(a^{-1})$$

and

$$1_H = \varphi(1_G) = \varphi(a^{-1} \times a) = \varphi(a^{-1})\varphi(a).$$

■

Proposition: Suppose G, H are groups. Then $G \cong H$ if and only if there exist homomorphisms $\varphi : G \to H$ and $\psi : H \to G$ such that $\varphi \circ \psi = \mathrm{id}_H$ and $\psi \circ \varphi = \mathrm{id}_G$. Now suppose the second statement holds. Then as functions, we see that φ, ψ are bijections, and their inverses are each other. Thus φ is a bijective homomorphism between G and H, and thus an isomorphism.

Proof: Suppose $G \cong H$. Thus there exists a bijection $\xi : G \to H$ that is a homomorphism. Then we can just take $\varphi = \xi$ and $\psi = \xi^{-1}$. ■

Definition (kernel): The *kernel* of a homomorphism $\varphi : G \to H$ is the set of elements in G that map to 1_H under φ. In other words, $\ker \varphi = \varphi^{-1}(\{1_H\})$. When $\ker \varphi = \{1_G\}$, we often call the kernel trivial.

Example *(kernels)*:
- For an isomorphism, the kernel will be $\{1_G\}$, as it's a bijection.
- The kernel of $\varphi : \mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$ where $\varphi(n) = n \pmod 3$ is $3\mathbb{Z}$.

Proposition: The kernel of a homomorphism $\varphi : G \to H$ is a subgroup of G, the image of a homomorphism is a subgroup of H, and $\varphi^{-1}(N)$ is a subgroup of G for $N \leq H$. Furthermore, a homomorphism is injective if and only if $\ker \varphi = \{1_H\}$.

Proof: Suppose $a, b \in \ker \varphi$. Then $\varphi(ab) = \varphi(a)\varphi(b) = 1_H$, so $ab \in \ker \varphi$. We already know that $1_G \in \ker \varphi$, and so is an identity for it. For any $a \in \ker \varphi$, we also have $1_H = \varphi(aa^{-1}) = \varphi(a)\varphi(a^{-1}) = \varphi(a^{-1})$, and thus $a^{-1} \in \ker \varphi$. Thus $\ker \varphi \leq G$.

Suppose $h_1, h_2 \in \varphi(G)$. Thus there exists $g_1, g_2 \in G$ such that $\varphi(g_1) = h_1, \varphi(g_2) = h_2$. We then have $\varphi(g_1 g_2) = \varphi(g_1)\varphi(g_2) = h_1 h_2$. Thus $h_1 h_2 \in \varphi(G)$. Note that $1_H \in \varphi(G)$ and so it has an identity. If $h \in \varphi(G)$ with $\varphi(g) = h$, then $1_H = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1}) = h\varphi(g^{-1})$. Similarly, we find $1_H = \varphi(g^{-1})h$. Thus $h^{-1} \in \varphi(G)$.

Since $1_H \in N$, we have $1_G \in \varphi^{-1}(N)$. If $g \in \varphi^{-1}(N)$, then $\varphi(g) \in N$. Then $\varphi(g)^{-1} \in N$. Since φ is a homomorphism, $\varphi(g)^{-1} = \varphi(g^{-1})$. Thus $g^{-1} \in \varphi^{-1}(N)$, so it contains inverses. If $a, b \in \varphi^{-1}(N)$, then $\varphi(ab) = \varphi(a)\varphi(b) \in N$, which implies $ab \in \varphi^{-1}(N)$, and so is closed. Thus $\varphi^{-1}(N)$ is a subgroup of G.

Now suppose φ is injective. We already know that $\varphi(1_G) = 1_H$, so by injectivity no other element of G can map to 1_H. Thus $\ker \varphi = \{1_G\}$. Now suppose the kernel is trivial, and suppose $\varphi(a) = \varphi(b)$. Then $\varphi(ab^{-1}) = \varphi(a)\varphi(b^{-1}) = \varphi(a)\varphi(b)^{-1} = 1_H$. Thus $ab^{-1} \in \ker \varphi$, which implies $ab^{-1} = 1_G \Rightarrow a = b$. Thus φ is injective. ■

1.1.2. Cosets and Quotient Groups

Definition (left/right coset, index): Let H be a subgroup of H. Then for any $g \in G$ the set

$$gH = \{gh : h \in H\}$$

is called a *left coset of* H. Similarly, the set

$$Hg = \{hg : h \in H\}$$

is called a *right coset of* H. The *index* of H in G, denoted $[G : H]$, is the number of left cosets of H in G.

Almost all statements we prove for left cosets will hold for right cosets, and vice versa. Note that $x \mapsto g_2 g_1^{-1} x$ is a bijection from $g_1 H$ and $g_2 H$, so the cosets of H have the same cardinality, in particular, $|H|$.

Proposition: Suppose $H \leq G$ and $g_1, g_2 \in G$. Then $g_1 H = g_2 H$ if and only if $g_1^{-1} g_2 \in H$.

Equivalently, the second condition can be $g_2^{-1} g_1$. The inverse element just has to be on the left.

Proof: Suppose $g_1 H = g_2 H$. Then $g_2 = g_2 e = g_1 h$ for some $h \in H$. Thus $g_1^{-1} g_2 = h \in H$. Now suppose $g_1^{-1} g_2 \in H$. Thus it's inverse $g_2^{-1} g_1 \in H$. Given $g_1 h \in g_1 H$, we have $g_1 H = g_2 (g_2^{-1} g_1) h \in g_2 H$, so $g_1 H \subseteq g_2 H$. Given $g_2 h \in g_2 H$, we have $g_1 (g_1^{-1} g_2) h \in g_1 H$, so $g_2 H \subseteq g_1 H$. ■

Note this implies that cosets are disjoint or equal, since if they has nonempty intersection, then $g_1 h_1 = g_2 h_2 \Rightarrow g_2^{-1} g_1 = h_2 h_1^{-1} \in H$ for some $h_1, h_2 \in H$, and by the above the two cosets are equal.

Theorem (Lagrange's theorem): Let G be a finite group and H a subgroup. Then $|H|$ divides $|G|$.

Proof: From the above, we know that distinct cosets are disjoint. Thus the distinct cosets of H partition G. Let k be the number of distinct cosets. Since cosets have the same size, this implies $k|H| = |G|$. ■

Corollary: If $x \in G$, where G is a finite group, then $x^{|G|} = 1$.

Proof: Let $H = \langle x \rangle \leq G$. By finiteness, ord $x = |H|$. Since $|H|$ divides $|G|$ by Lagrange's theorem, we have $x^{|G|} = \left(x^{|H|}\right)^{|G|/|H|} = 1$, as desired. ■

Proposition (only prime order group is cyclic): Let p be a prime. Every order p group is isomorphic to $\mathbb{Z}/p\mathbb{Z}$.

Proof: Suppose $|G| = p$. Note that every element can either have order 1 or p by the above corollary. If an element has order 1, then by definition it's the identity. Thus every other element in G has order p. Thus we can pick $g \in G$ (namely any element other than the identity) such that $\{g^0, g, g^2, g^3, \ldots, g^{p-1}\} = G$. Now define $\varphi : G \to \mathbb{Z}/p\mathbb{Z}$ by $\varphi(g^k) = k \pmod p$. Thus φ is a bijection, and

$$\varphi(g^a g^b) = \varphi(g^{a+b}) = a + b \pmod p = \varphi(g^a) + \varphi(g^b) \pmod p.$$

Thus φ is an isomorphism, so G and $\mathbb{Z}/\ p\mathbb{Z}$ are isomorphic. ■

Proposition: Let H, K be finite subgroups of a group G, and let

$$HK = \{hk : h \in H, k \in K\}.$$

Then

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

Proof: Note that hK is a left coset of K. Since cosets are non-overlapping, we just need to count the number of distinct H left cosets of K. From above, we know that $h_1K = h_2K$ if and only if $h_1^{-1}h_2 \in K$. Let $h_1 \sim h_2$ if $h_1^{-1}h_2 \in K$. It's easy to see that this is an equivalence relation on K, and so the equivalence classes form a partition. We claim that the equivalence classes are the cosets of $H \cap K$ in H.

Consider some $hH \cap K$. First we show that these all lie in the same equivalence class. Let $hh_1, hh_2 \in HH \cap K$. Then $(hh_1)^{-1}hh_2 = h_1^{-1}h_2 \in K$, so $hh_1 \sim hh_2$.

Now we show that if two elements are equivalent, they must lie in the same coset. Since $H \cap K \leq H$, the cosets partition H. Thus for $h_1, h_2 \in H$, there exist $h_1', h_2' \in H$ for which $h_1 \in h_1'H \cap K$ and $h_2 \in h_2'H \cap K$. Thus we can write

$$h_1 = h_1'h_1'' \quad \text{and} \quad h_2 = h_2'h_2''$$

for $h_1'', h_2'' \in H \cap K$. From the equivalence, we have $h_1^{-1}h_2 \in K$. Thus

$$h_1''^{-1}h_1'^{-1}h_2'h_2'' \in K.$$

Note that each lie in H as well, so the left lies in $H \cap K$. Since the outer most elements also lie in $H \cap K$, we see that $h_1'^{-1}h_2' \in H \cap K$. Thus $h_1'H \cap K = h_2'H \cap K$. Thus the equivalence classes are the cosets, as desired.

There are $\frac{|H|}{|H \cap K|}$ such cosets. Since each left coset of K has size $|K|$, we thus have

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

■

To motivate quotient groups, suppose $\varphi : G \to Q$ is a surjective homomorphism, and suppose $h \in \ker \varphi$. Then note for any $g \in G$, we have $\varphi(gh) = \varphi(g)\varphi(h) = \varphi(g)$. Thus φ basically ignores elements of $\ker \varphi$. Roughly speaking, this means Q is isomorphic to G when we "mod out" by $\ker \varphi$.

As an example, take $\varphi : \mathbb{Z} \to \mathbb{Z}/10\mathbb{Z}$ where $\varphi(n) = n \pmod{10}$. Then when we "mod out" but multiples of 10 in $\mathbb{Z}$, we get all the elements of $\mathbb{Z}/10\mathbb{Z}$, i.e. $\{\ldots, -20, -10, 0, 10, 20, \ldots\} \to 0, \{\ldots, -19, -9, 1, 11, 21, \ldots\} \to 1$, etc. Thus we can think of Q as the quotient of G and $\ker \varphi$, with the elements of it being cosets of $\ker \varphi$. We now make all of this precise.

Definition (normal): A subgroup N of G is called *normal* if it is the kernel of some homomorphism. We write $N \trianglelefteq G$.

Definition (quotient group): Let $N \trianglelefteq G$. Then the *quotient group*, denote G/N, is group with elements being left cosets of N. The operation is given by taking an elements $C_1, C_2 \in G/N$, picking $g_1 \in C_1, g_2 \in C_2$, and letting C_1C_2 being the coset containing g_1g_2.

Note that the operation is equivalent to declaring $(g_1N) \cdot (g_2N) = (g_1g_2)N$, which is often how we'll treat the operation.

We need to check that this operation is well-defined, and that it actually forms a group. First suppose $a \in g_1N$ and $b \in g_2N$. Thus $a = g_1h_1, b = g_2h_2$ for some $h_1, h_2 \in N$. We need to show that $abN = g_1g_2N$, which is equivalent to showing $g_2^{-1}g_1^{-1}ab \in N$. Since N is normal, there exists some homomorphism φ into some group such that $\ker \varphi = N$. Then we have

$$\varphi(g_2^{-1}g_1^{-1}ab) = \varphi(g_2^{-1})\varphi(g_1^{-1})\varphi(g_1)\varphi(h_1)\varphi(g_2)\varphi(h_2) = \varphi(g_2^{-1})\varphi(g_1^{-1})\varphi(g_1)\varphi(g_2) = 1.$$

Thus $g_2^{-1}g_1^{-2}ab \in N$, so this operation is well-defined. It's easy to see that 1_GN is the identity, and that $g^{-1}N$ is the inverse of gN. Thus G/N is indeed a group.

Now we'd like a way to identify normal subgroups without having to to reference a homorphism. It turns out this is the condition we'd like:

Proposition (normality condition): Let $H \leq G$. Then $H \trianglelefteq G$ if and only if for all $g \in G$ and $h \in G$, $ghg^{-1} \in H$. We often write the second part as $gHg^{-1} = H$ for all $g \in G$.

Proof: First suppose $H \trianglelefteq G$. Thus there exists some group K and homomorphism $\varphi : G \to K$ such that $\ker \varphi = H$. Then for any $g \in G, h \in H$, we have $\varphi(ghg^{-1}) = \varphi(g)\varphi(h)\varphi(g^{-1}) = \varphi(g)\varphi(g^{-1}) = 1_K$, so $ghg^{-1} \in H$.

Now suppose $gHg^{-1} = H$ for all $g \in G$. Let $[G/H]$ contain all cosets of H, and define the operation as $(g_1H)(g_2H) = (g_1g_2)H$. To show this is well defined, suppose $a = g_1h_1$ and $b = g_2h_2$ for some $h_1, h_2 \in H$. We need to show that $(ab)H = (g_1g_2)H$, which means we just need to show that $g_2^{-1}g_1^{-1}ab \in H$. To do this, note that

$$g_2^{-1}g_1^{-1}ab = g_2^{-1}g_1^{-1}g_1h_1g_2h_2 = g_2^{-1}h_1g_2h_2.$$

Then by the condition, there exists $h_3 \in H$ such that $g_2^{-1}h_2g_2 = h_3$. Thus

$$g_2^{-1}g_1^{-1}ab = h_3h_2 \in H$$

Thus the cosets are the same, and the multiplication is well-defined. The identity is 1_GH, and the inverse of gH is $g^{-1}H$.

Now let $\varphi : G \to [G/H]$ be defined by $\varphi(g) = gH$. We have $\varphi(g_1g_2) = (g_1g_2)H = (g_1H)(g_2H) = \varphi(g_1)\varphi(g_2)$, so φ is a homorphism. If $h \in H$, then $\varphi(h) = hH = H$, so $H \subseteq \ker \varphi$. If $g \in \ker \varphi$, then $gH = H$, which implies that $g \in H$. Thus $\ker \varphi \subseteq H$. Since we've constructed a homomorphism with kernel H, it is indeed a normal subgroup. ■

Finally, we have another formal way of saying that G/N "acts" like the group a surjective homomorphism with kernel N sends G to.

Theorem (first isomorphism theorem): Let $\varphi : G \to H$ be a homomorphism. Then $G/\ker\varphi \cong \varphi(G)$.

Proof: Let $N = \ker\varphi$. Define $\psi : G/N \to \varphi(G)$ by $\psi(gN) = \varphi(g)$. Note this is well defined since if a, b represent the same coset, then they must have the same φ value (as $a^{-1}b \in H \Rightarrow \varphi(a^{-1}b) = 1_H$). We have $\psi((g_1g_2)N) = \varphi(g_1g_2) = \varphi(g_1)\varphi(g_2) = \psi(g_1N)\psi(g_2N)$. Thus ψ is a homomorphism. Note if that $\psi(gN) = \varphi(g) = 1_H$, then $g \in N$. Thus $\ker\psi = \{1_GN\}$, so ψ is injective. Surjectivity is easy, since $\varphi(g) = \varphi(gN)$ for all $g \in G$. ■

Theorem (fourth isomorphism theorem): Let $\varphi : G \to G'$ be a surjective homomorphism. There exists a natural bijection between

$$\{\text{subgroups of } G \text{ containing } \ker\varphi\} \longleftrightarrow \{\text{subgroups of } G'\}.$$

Given a subgroup $H \leq G$ with contains $\ker\varphi$, it maps to $\varphi(H) \leq G'$.

Proof: Given a subgroup $H \leq G$ containing $\ker\varphi$, we show that $\varphi^{-1}(\varphi(H)) = H$, and given $H' \leq G'$, we show that $\varphi(\varphi^{-1}(H')) = H'$.

First suppose $H' \leq G'$. From basic set theory, we have $\varphi(\varphi^{-1}(H')) \subseteq H'$. Now suppose $h' \in H'$. Thus there exists $g \in G$ such that $\varphi(g) = h'$. Thus $g \in \varphi^{-1}(H')$, which implies $h' \in \varphi(\varphi^{-1}(H'))$, so we have the other inclusion.

Now suppose $H \leq G$ contaning $\ker\varphi$. Again from basic set theory, $\varphi^{-1}(\varphi(H)) \supseteq H$. Now suppose $g \in \varphi^{-1}(\varphi(H))$. Thus there exists $f \in H$ such that $\varphi(g) = \varphi(f) \Rightarrow \varphi(gf^{-1}) = 1$. Thus $gf^{-1} \in \ker\varphi \leq H$. Since $f \in H$, this implies $g \in H$, so $\varphi^{-1}(\varphi(H)) \subseteq H$. ■

1.2. Group Actions

Definition (group action): Let X be a set and G be a group. A *group action* is a binary operation $\cdot : G \times X \to X$, which lets g send x to $g \cdot x$. This operation satisfies the following:
- $(g_1g_2) \cdot x = g_1 \cdot (g_2 \cdot x)$ for all $g_1, g_2 \in G, x \in X$.
- $1_G \cdot x = x$ for all $x \in X$.

Example: Consider $G = D_8$, and let it act on the set of colorings of the vertices of a square. If a square x has tl and br red and tr and bl blue, then $r^2 \cdot x = x$.

Definition (orbit): Given a group action G on X, we define an *orbit* to be an equivalence class on X, where $x \sim y$ if there exists $g \in G$ such that $y = g \cdot x$.

It's easy to see that the orbits partition G.

Definition (stabilizer): The *stabilizer* of a point $x \in X$, denote $\mathrm{Stab}_G(x)$, is the set of $g \in G$ such that $g \cdot x = x$.

It's easy to see that $\mathrm{Stab}_G(x)$ is a subgroup of G.

Theorem (orbit-stabilizer theorem): Let $\mathcal{O}$ be an orbit, and pick any $x \in \mathcal{O}$. Let $S = \mathrm{Stab}_G(x)$. Then

$$|\mathcal{O}||S| = |G|.$$

Proof: Let C be the set of cosets of S (which by Lagrange's theorem we know has size $|G|/|S|$), and define $f : C \to \mathcal{O}$ given by $f(gS) = g \cdot x$. This is well defined, since if $g_1 S = g_2 S$, we have $g_1^{-1} g_2 \in S$, so $g_1^{-1} g_2 \cdot x = x \Rightarrow g_2 \cdot x = g_1 \cdot x$. Reversing this logic also shows that f is injective. Surjectivity follows since every element of $\mathcal{O}$ takes the form $g \times x$. Thus f is a bijection, implying

$$|\mathcal{O}| = |C| = |G|/|S|,$$

as desired. ■

In particular, this result implies that the stabilizers of elements in an orbit have the same size.

Definition (fixed points): Let G act on X. Then for $g \in G$, let

$$\mathrm{FixPt}\ g = \{x \in X : g \cdot x = x\}.$$

Lemma (Burnside's lemma): Let G act on X. Let B be the set of orbits in X. Then

$$|B| = \frac{1}{|G|} \sum_{g \in G} |\mathrm{FixPt}\ g|.$$

Proof: We double count the number of pairs $(g, x) \in G \times X$ such that $g \cdot x = x$. Let N be that number. If we fix g, then the number of x that work is $|\mathrm{FixPt}\ g|$. Thus

$$N = \sum_{g \in G} |\mathrm{FixPt}\ g|.$$

If we fix x instead, the number of g that work is $|\mathrm{Stab}_G(x)|$, so

$$\begin{aligned} N = \sum_{g \in G} |\mathrm{FixPt}\ g| &= \sum_{x \in X} |\mathrm{Stab}_G(x)| \\ &= \sum_{\mathcal{O} \in B} \sum_{x \in \mathcal{O}} |G|/|\mathcal{O}| \\ &= \sum_{\mathcal{O} \in B} |G| \\ &= |G||B|, \end{aligned}$$

where the third equality follows from the orbit-stabilizer theorem. Dividing by $|G|$ yields the desired conclusion. ■

One particular type of group action is especially important.

Definition (conjugation): Let G act on itself as follows:

$$g : h \mapsto ghg^{-1}.$$

This is called *conjugation*. The *conjugacy classes* of a group G are the orbits of G under the conjugacy action.

Definition (center): The *center* of a group G, denoted $Z(G)$, is the set of elements $x \in G$ such that $xg = gx$ for all $g \in G$.

It's easy to see that $Z(G)$ is normal, and that each element in $Z(G)$ is its own conjugacy class.

Definition (centralizer): Suppose $s \in G$. Let $C_G(s)$ denote the *centralizer* of s, which defined to be the stabilizer of s under conjugation. In other words,

$$C_G(s) = \{g \in G : gsg^{-1} = s\}.$$

Proposition (class equation): Let G be a finite group. Show that

$$|G| = |Z(G)| + \sum_{s \in S} \frac{|G|}{|C_G(s)|},$$

where $S \subseteq G$ is defined as follows: for each conjugacy class $C \subseteq S$ with $|C| > 1$, pick a representative of C and add it to S. Furthermore, each term of the above sum divides $|G|$.

Proof: Let $\mathcal{O}_1, ..., \mathcal{O}_m$ be the orbits of G under conjugation that have size greater than 1. If $s \in Z(G)$, then clearly its orbit size is one. Similarly, if s has orbit size one, then it commutes with every element, and so lies in $Z(G)$. Thus the $\mathcal{O}_i$ and $Z(G)$ partition G, so

$$|G| = |Z(G)| + \sum_{k=1}^{m} |\mathcal{O}_k|.$$

Since $Z(G) \leq G$, its order clearly divides $|G|$. By orbit-stabilizer, each of the terms of the sume divides $|G|$. Furthermore, by orbit-stabilizer, we can replace $|\mathcal{O}_k|$ with $\frac{|G|}{|C_G(s)|}$, where we pick some $s \in \mathcal{O}_k$. Thus we can the desired equation and divisibility. ■

We can also define conjugation naturally on subgroups, by letting g act on $H \leq G$ by $H \mapsto gHg^{-1}$. Thus two subgroups $H, K \leq G$ are *conjugate subgroups* if there exists $g \in G$ such that $K = gHg^{-1}$. Note that conjugation is an isomorphism, so conjugate subgroups are isomorphic. Also note that normal subgroups are fixed by all $g \in G$.

Definition (normalizer): For $H \leq G$, the *normalizer* of H is

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

This is just the stabilizer of H under the conjugation action.

1.3. Sylow Theorems

Theorem (Sylow theorems): Let G be a group of order $p^n m$, where $\gcd(p, m) = 1$ and p is prime. A *Sylow p-subgroup* is a subgroup of order p^n. Let n_p be the number of Sylow p-subgroups.

a) $n_p \equiv 1 \pmod{p}$. In particular, $n_p \neq 0$ and a Sylow p-subgroup exists.
b) $n_p \mid m$.
c) Every p-group is contained in a Sylow p-subgroup. In particular, any two Sylow p-subgroups are conjugate subgroups, and thus isomorphic.

Proof: First we show that a Sylow p-subgroup exists. Let X be the set of all size p^n subsets of G, and let G act on X by left multiplication: $g \cdot S = \{gs : s \in S\}$. We claim there exists an orbit $\mathcal{O}$ under this action with size not divisible by p. To show this, note that

$$\begin{aligned}
\nu_p\left(\binom{p^n m}{p^n}\right) &= \nu_p\left(\frac{(p^n m)!}{(p^n)!(p^n(m-1))!}\right) \\
&= \nu_p((p^n m)!) - \nu_p((p^n)!) - \nu_p((p^n(m-1))!) \\
&= \frac{p^n - 1}{p - 1}(m - 1 - (m - 1)) \\
&= 0.
\end{aligned}$$

Thus $|X| = \binom{p^n m}{p^n} \neq 0 \pmod{p}$. Since the orbits partition X, this implies there exists an orbit not divisible by p, as desired.

Now pick $S \in \mathcal{O}$ and let $H = \mathrm{Stab}_G(S)$. By the orbit-stabilizer theorem, we have $|\mathcal{O}||H| = |G| = p^n m$. Since p^n doesn't divide $|\mathcal{O}|$, it must divide $|H|$. Now consider a second action where we let H act on S by left multiplication (which we can do since $H = \mathrm{Stab}_G(S)$, so $hs \in S$). Now pick $s \in S$, and consider $\mathrm{Stab}_H(s)$. If $h \in \mathrm{Stab}_H(s)$, then $hs = s \Rightarrow h = 1_H$. Thus $\mathrm{Stab}_H(s) = \{1_H\}$. Thus by the orbit-stabilizer theorem, all orbits of this action have size $|H|$. Thus $|H|$ divides $|S| = p^n$. Thus $|H| = p^n$, so we have a Sylow p-subgroup.

Next we show that any two Sylow p-subgroups are conjugate. Let $Q \leq G$ have order a power of p, and let P be a Sylow p-subgroup. Let Q act on the left cosets of P by left multiplication. Since Q has order power of p, by the orbit-stabilizer theorem, every orbit has size divisible by p or is 1. There are m left cosets of P and m is not divisible by p. Since the orbits partition the cosets, there must exist some orbit with size 1. Thus there exists some g for which $qgP = gP$ for all $q \in Q$. Thus $qg \in gP$ for all $q \in Q$, which implies $q \in gPg^{-1}$ in for all $q \in Q$. Thus $Q \subseteq gPg^{-1}$. Then if Q is a Sylow p-subgroup, it has size p^n, and so the sets must be equal. Thus any two Sylow p-subgroups are conjugate. This process also proves that every p-group lies in some Sylow p-subgroup.

Now we show that $n_p \equiv 1 \pmod{p}$. Let $\mathcal{S}$ be the set of all Sylow p subgroups, which by definition has size n_p. From the above, $\mathcal{S}$ is nonempty with some element P. Let P act on $\mathcal{S}$ by conjugation. Again since P has order power of p, the orbits have size power of p or 1. Thus the number of fixed points equivalent to n_p modulo p.

Clearly P is a fixed point, but we show that it's the only fixed point. To do so, suppose Q is a fixed point, meaning $xQx^{-1} = Q$ for all $x \in P$. Let $N_G(Q)$ be the normalizer of Q. It obviously contains Q, and since Q is a fixed point of P under it's action, the normalizer also contains P. Since $N_G(Q) \leq G$, it has order that divides $p^n m$. Since it contains P, it must have size at least p^n. Thus P, Q are Sylow p-subgroups of $N_G(Q)$. Then by c) of the theorem, we see that P and Q are conjugate subgroups of $N_G(Q)$. Thus there exists $g \in N_G(Q)$ for which $gQg^{-1} = P$. But by definition of being in the normalizer, we see that $gQg^{-1} = Q$. Thus $Q = P$, as desired.

Finally, we show that $n_p \mid m$. Since we showed that $p \nmid n_p$, we can just show that n_p divides $|G|$. Let G act on the set of all Sylow p-subgrous by conjugation. By c), we see that there's only one orbit under this action, so the conclusion follows from the orbit-stabilizer theorem. ■

Proposition (some consequences of Sylow theorems):
a) A Sylow p-subgroup is normal if and only if $n_p = 1$.
b) If G has order pq with $p < q$, then it has normal subgroup of order q.
c) If p is prime and divides the order of an abelian group, then there is exactly one Sylow p-subgroup.
d) (Cauchy's theorem) If p divides the order of a group, then there exists an element with order p.

Proof:
a) If H is a Sylow p-subgroup of G and $H \trianglelefteq G$. Then for all $g \in G$, we have $gHg^{-1} = H$. However, any two Sylow p-subgroups are conjugate, so the only Sylow p-subgroup is H. Thus $n_p = 1$. If $n_p = 1$, then since any two Sylow p-subgroups are conjugate, we must have $gHg^{-1} = H$ for all $g \in G$, and thus $H \trianglelefteq G$.
b) From the Sylow theorems, we see that $n_q \mid p$. Since $n_q = kq + 1$ for some integer k, we see that n_q must be 1. Then by a), we have the desired conclusion.
c) Conjugation is the identity map in an abelian group, so again there can only be one Sylow p -subgroup.
d) By the Sylow theorems, there exists a subgroup H of order p^n. Pick some nonidentity element $g \in H$. Then by Lagrange's theorem, the group $\langle g \rangle$ has order p^t with $1 \leq t \leq n$. If $t = 1$, we're done. Otherwise take the element $g^{p^{t-1}}$. It's easy to see this has order p.

■

1.4. Problems

Problem (Cayley's theorem): Let G be a finite group. Show that there exists n such that G is isomorphic to some subgroup of S_n.

Solution: Let $G = \{g_1, g_2, ..., g_n\}$ be the elements, with g_1 being the identity. Given an element $g \in G$, let $\text{ind}(g)$ denote the index of it's place in G by the order in the set set given above. Thus $\text{ind}(g_i) = i$. Now define $\varphi : G \to S_n$ as

$$\varphi(g_i) = \{\text{ind}(g_i g_1), ..., \text{ind}(g_i g_n)\}.$$

Now we verify that φ is a homomorphism. Then we know that $\varphi(G) \leq S_n$, and then by definition G and the image are isomorphic.

We have that

$$\varphi(g_i g_j) = \{\text{ind}(g_i g_j g_1), ..., \text{ind}(g_i g_j g_n)\}.$$

Pick $k \in \{1, ..., n\}$. Then

$$(\varphi(g_i)\varphi(g_j))(k) = \varphi(g_i)(\text{ind}(g_j g_k)) = \text{ind}\left(g_i g_{\text{ind}(g_j g_k)}\right) = \text{ind}(g_i g_j g_k) = \varphi(g_i g_j)(k).$$

Thus $\varphi(g_i g_j) = \varphi(g_i)\varphi(g_j)$, as desired.

Problem: Let G be a finite abelian group. Prove that for all $x \in G$, we have $x^{|G|} = 1$ without appealing to Lagrange's theorem.

Solution: Fix $x \in G$, and let $G = \{g_1, g_2, ..., g_n\}$. Then by the cancellation law, $\{xg_1, xg_2, ..., xg_n\} = G$. Thus

$$\prod_{i=1}^{n} xg_i = \prod_{i=1}^{n} g_i \Rightarrow x^n = 1,$$

as desired.

Problem: Find all groups G such that $\varphi : G \to G$ defined by $\varphi(g) = g^2$ is a homomorphism.

Solution: Suppose G is abelian. Then $\varphi(ab) = (ab)^2 = a^2b^2 = \varphi(a)\varphi(b)$. Now suppose φ is a homorphism. Then $\varphi(ab) = abab = a^2b^2 = \varphi(a)\varphi(b) \Rightarrow ba = ab$, and thus φ is abelian.

Problem: Let G and H be finite groups, where $|G| = 1000$ and $|H| = 999$. Show that a homomorphism $G \to H$ must be trivial.

Solution: Suppose $\varphi : G \to H$ is a homomorphism. By the first isomorphism, $G/\ker\varphi \cong \varphi(G)$. Note that $|G/\ker\varphi| = |G|/\ |\ker\varphi|$, as the number of cosets of $\ker\varphi$ is the number of elements in $G/\ker\varphi$, and from the proof of Lagrange's theorem, we know that a subgroup N produces $|G|/|N|$ cosets. Thus

$$|G|/|\ker\varphi| = |\varphi(G)|.$$

Since $\varphi(G) \leq H$, by Lagrange's theorem, $|\varphi(G)|$ divides $|H|$. Now if $|\ker \varphi| \neq |G|$, then the left side above contains a factor of 2 or 5, while the right side contains contains factors of 3 or 37. Thus we must have $|\ker \varphi| = |G|$, which implies the homomorphism is trivial.

Problem: Show that two elements in the same conjugacy class have the same order.

Solution: Suppose $a, b \in G$ are conjugate. Thus there exists $g \in G$ such taht $gag^{-1} = b$. Then

$$b^n = (gag^{-1})^n = \underbrace{gag^{-1} \cdots gag^{-1}}_{n \text{ times}} = ga^n g^{-1}.$$

Thus if $\operatorname{ord} a = n$, then $b^n = ga^n g^{-1} = gg^{-1} = 1$. If $\operatorname{ord} b = n$, then $b^n = 1 = ga^n g^{-1} \Rightarrow g = ga^n \Rightarrow a^n = 1$. Thus they have the same order, so we're done.

Problem: Assume G is a finite group of order $n \geq 2$ and p is the smallest prime dividing n. Let H be a subgroup of G with $|G|/|H| = p$. Show that H is normal in G.

Solution: Let H act on the left cosets of H by left multiplication. Pick some left coset gH and let $\mathcal{O}$ be it's orbit. By the orbit-stabilizer theorem, $|\mathcal{O}|$ divides $|H|$, which divides n. However, $|\mathcal{O}| \leq p$, since there are p cosets of H by $|G|/|H| = p$. Since p is the smallest prime dividing $|G|$, this forces either $|\mathcal{O}| = 1, p$.

First suppose $|\mathcal{O}| = p$. Then by the orbit-stabilizer theorem, $|\mathcal{O}||\mathrm{Stab}_H(H)| = |H|$ (as $\mathcal{O}$ contains all cosets). However, since $\mathrm{Stab}_H(H) = H$, it has size at $|H|$, which implies $|\mathcal{O}| = 1$, contradiction.

Thus $\mathcal{O} = \{gH\}$. We can do this for all cosets and see each is a fixed point. Thus for any $h \in H$, we have $hgH = gH$, which implies $g^{-1}hg \in H$. Thus $H \trianglelefteq G$.

1.5. Classifying Groups of Order n

This section will be dedicated to classifying all groups of a certain order (up to isomorphism), for reasonable n. Before the classification, we gather results that will be helpful, some from earlier and some new.

Proposition: Every group of prime order is cyclic.

Proposition: Every p-group has nontrivial center.

Proposition: The groups of order p^2 are $C_p \times C_p$ and C_{p^2}, where p is prime.

Proof: We show a group of order p^2 must be abelian, from which the result follows using the fundamental theorem of finitely generated abelian groups. From the previous proposition, we know that $Z(G)$ is not trivial. Suppose $|Z(G)| = p$, and pick $g \notin Z(G)$. From the class equation, we know that $|\mathcal{O}_g|$ has to divide p^2 and must be at least p. It obivously can't be equal to p^2, so $|\mathcal{O}_g| = p \Rightarrow |C_G(g)| = p$. Since $Z(G) \leq C_G(g)$, we thus have $Z(G) = C_G(g) \Rightarrow g \in Z(G)$, contradiction. Thus $|Z(G)| = p^2 \Rightarrow Z(G) = G$, as desired. ■

Proposition: Every group of order $2p$ is either cyclic or dihedral.

Proposition (product group stuff): Let H and K be subgroups of a group G, and let $\psi : H \times K \to G$ be the map $\psi(h, k) = hk$. It's image is the set $HK = \{hk : h \in H, k \in K\}$.

a) ψ is injective if and only if $H \cap K = \{1\}$.
b) ψ is a homomorphism from $H \times K$ to G if and only if elements of K commute with elements of H.
c) If $H \trianglelefteq G$, then $HK \leq G$.
d) ψ is an isomorphism if and only if $H \cap K = \{1\}$, $HK = G$, and both H and K are normal subgroups of G.

Proof:

a) First suppose $H \cap K = \{1\}$, and suppose $\psi(h, k) = hk = h'k' = \psi(h', k')$. From this, we have $h'^{-1}h = k'k^{-1}$. Thus both sides lie in $H \cap K$, and thus must be 1. Thus $h = h', k = k'$, so ψ is injective. Now suppose ψ is injective. If $h \in H \cap K$, then so is h^{-1}. Thus $\psi(h, h^{-1}) = 1 = \psi(1, 1)$, so by injectivity, we must have $h = 1$, as desired.

b) Suppose H and K commute. Then

$$\psi(h, k)\psi(h', k') = hkh'k' = hh'kk' = \psi(hh', kk').$$

Now suppose ψ is a homomorphism. Cancelling the left and right elements from the above calculation, we see $kh' = h'k$.

c) We have

$$hkh'k' = h(kh')k' = hh''kk' \in HK,$$

where the second equality follows from normality and $kh'k^{-1} = h''$. Invertibility follows similarly.

d) First we show the backwards direction. This follows from a), b), and c). Now suppose ψ is an isomorphism. Then by a), $H \cap K = \{1\}$. From surjectivity, we also have $HK = G$. By b), elements of H and K must commute. For any $g \in G$, write $g = hk$ for the unique $(h, k) \in H \times K$ from ψ. Then for $h' \in H$, we have

$$gh'g^{-1} = hkh'k^{-1}h^{-1} = hh'h^{-1} \in H.$$

Thus $H \trianglelefteq G$. A similar process shows that K is normal in G.

■

Proposition: $G/Z(G)$ is cyclic if and only if G is abelian.

In particular, the above implies the index of $Z(G)$ in G can never be prime.

Theorem (Sylow theorems):

Theorem (fundamental theorem of finitely generated abelian groups/ FTFGAG):

See module section for proof.

Theorem: abelian forcing numbers

Order 1

Let $G = \{e\}$. Any operation $\times : G \times G \to G$ will just be $e \times e = e$, so only one group exists.

Order 2

Since 2 is prime, the only group is C_2.

Order 3

Since 3 is prime, the only group is C_3.

Order 4

Since $4 = 2^2$, any group must be abelian. Then by FTFGAG, a group of order 4 is either C_4 or $C_2 \times C_2$.

Order 5

Since 5 is prime, the only group is C_5.

Order 6

Since $6 = 2 \cdot p$, the only groups are C_6 and $D_6 \cong S_6$.

Order 7

Since 7 is prime, the only group is C_7.

Order 8

Order 9

Since $9 = 3^2$, the only groups are C_9 and $C_3 \times C_3$.

Order 10

Since $10 = 2 \cdot 5$, the only groups and C_{10} and D_{10}.

Order 11

Since 11 is prime, the only group is C_{11}.

Order 12

Order 13

Since 13 is prime, the only group is C_{13}.

Order 14

Since $14 = 2 \cdot 7$, the only groups are C_{14} and D_{14}.

2. Rings and Modules

2.1. Ring Basics

Definition (ring): A *ring* is a triple $(R, +, \times)$ such that the following hold:
- $(R, +)$ is an abelian group, with identity 0_R, often denoted just 0.
- $\times$ is an associative, binary operation on R.
- Multiiplication distributes over addition.

The ring R is *commutative* if $\times$ is commutative. It is a *ring with identity* if there is an element $1_R \in R$ such that $1_R \times a = a \times 1_R = a$ for all $a \in R$.

Unless otherwise specified, after this section we will be working with rings with identity. We will also often be dealing with commutative rings.

Example:
- $\mathbb{Z}$ is basically the canonical example of a ring, as it has many desirable properties we will defined soon, and its the easiest to understand.
- Given a commutative group, denote the operator as $+$. Then define $a \times b = 0$ for any a, b in the group. This is called a *trivial ring* since the multiplication adds no new structure. Note also that this is a ring without identity.

a) The *trivial ring* $R = \{0\}$ has identity $0 = 1$. We will often exclude this ring by specifying $0 \neq 1$ (or just ignoring it).
b) The ring of polynomials $\mathbb{R}[x]$ with real coefficients, with addition and multiplication defined as expected.
c) Given some element $\alpha \in \mathbb{C}$, let $\mathbb{Q}[\alpha]$ be the ring of formal polynomials in α. If α is algebraic, then α^N can be written as the sum of smaller powers for some N, so in this case $\mathbb{Q}[\alpha]$ is just a set of finite polynomials in α. For example, $\alpha = \sqrt{-5}$ yields $\mathbb{Q}\left[\sqrt{-5}\right] = \left\{a + b\sqrt{-5} : a, b \in \mathbb{Q}\right\}$ and $\alpha = i$ yields $\mathbb{Q}[i] = \{a + bi : a, b \in \mathbb{Q}\}$.
d) The group $\mathbb{Z}/n\mathbb{Z}$ with multiplication defined as usual is a ring.

Definition (division ring/field): A ring R with identity $1 \neq 0$ is called a *division ring* if every nonzero element $a \in R$ has a multiplicative inverse, i.e. there exists $b \in R$ such that $ab = ba = 1$. A commutative division ring is called a *field*.

Example *(fields)*: Typical fields include $\mathbb{Q}, \mathbb{R}, \mathbb{C}$, and $\mathbb{F}_p$, where p is prime. Another field is the set of algebraic numbers.

Proposition (ring operation properties): Let R be a ring.

a) $0 \times a = a \times 0 = 0$.
b) $(-a) \times b = a \times (-b) = -(a \times b)$ for all $a, b \in R$.
c) $(-a) \times (-b) = a \times b$ for all $a, b \in R$.
d) If R has identity 1, then the identity is unique and $-a = (-1) \times a = a \times (-1)$.

Proof:
a) $0 \times a = (0 + 0) \times a = 0 \times a + 0 \times a \Rightarrow 0 \times a = 0$. The other equality follows similarly.
b) $a \times b + (-a) \times b = (a + (-a)) \times b = 0 \times b = 0 \Rightarrow (-a) \times b = -(a \times b)$. The other direction follows similarly.
c) By b), we have $(-a) \times (-b) = -(a \times (-b)) = -(-(a \times b)) = a \times b$, and the other direction follows similarly.
d) Follows from b) and the definition of identity.

■

Definition (zero divisor/unit): Let R be a ring. A nonzero element $a \in R$ is called a *zero divisor* if there is a nonzero element $b \in R$ such that $ab = 0$ or $ba = 0$. If R has identity $1 \neq 0$, an element $u \in R$ is called a *unit* if there exists $v \in R$ such that $uv = vu = 1$. The set of units is denoted $R^\times$.

It's easy to see that zero divisors can't be units, and so fields have no zero divisors. From this we can also define a field to be a commutative ring where every nonzero element is a unit.

Definition (integral domain): A commutative ring with identity $1 \neq 0$ is called *integral domain* if it has no zero divisors.

Example: $\mathbb{Z}/10\mathbb{Z}$ is not integral since $5 \cdot 2 \equiv 0 \pmod{10}$.

Proposition (cancellation law): Suppose $a, b, c \in R$ such that a is not a zero divisor. If $ab = ac$, then either $a = 0$ or $b = c$.

Proof: Subtracting yields $ab - ac = 0 \Rightarrow a(b - c) = 0$. If $a \neq 0$, then since it's not a zero divisor, there exists no nonzero element d such that $ad = 0$. Thus we must have $b - c = 0$. ■

Proposition: Any finite integral domain is a field.

Proof: Let R be a finite integral domain, and pick nonzero $a \in R$. Then by the cancellation law, the map $x \mapsto ax$ is injective. Since R is finite this map is also surjective. Thus there exists $b \in R$ such that $ab = 1$. Since a was arbitrary, there also exists $c \in R$ such that $bc = 1$. Thus b has

a left and right inverse. Since the set of units is a group, this implies b is invertible, with $a = c$ being its inverse. Again since a was arbitrary, this holds for every $r \in R$. Thus R is a field. ■

Definition (subring): A *subring* of a ring R is a subgroup of R that is closed under multiplication.

2.1.1. Homomorphisms, Quotient Rings, and Ideals

Definition (homomorphism/isomorphism): A *ring homomorphism* between rings R and S is a map $\varphi : R \to S$ such that

$$\varphi(a+b) = \varphi(a) + \varphi(b) \text{ and } \varphi(ab) = \varphi(a)\varphi(b)$$

for all $a, b \in R$ and such that $\varphi(1_R) = \varphi(1_S)$. If φ is bijective, then φ is a *ring isomorphism*, and R and S are *isomorphic*, denoted $R \cong S$.

When context is clear, we'll just say R and S are isomorphic.

Definition (kernel): If $\varphi : R \to S$, the set of elements of R that maps to 0 in S is called the *kernel of* φ, and denoted $\ker \varphi$.

Simalar to groups, it's easy to show that a homomorphism is injective if and only if its kernel only contains 0.

Example *(ring homorphisms)*:
- The map $\varphi : \mathbb{Z} \to 0$, where 0 denotes the zero ring, is a ring homomorphism.
- The map $\varphi : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ where $\varphi(n) = n \pmod 2$ is a ring homomorphism.

Proposition: Let $\varphi : R \to S$ be a homomorphism. Then $\varphi(R)$ is a subring of S. $\ker \varphi$ is also a subring of R, and if $\alpha \in \ker \varphi$, then $r\alpha, \alpha r \in \ker \varphi$ for all $r \in R$. Furthermore, if T is a subring of S, then $\varphi^{-1}(T)$ is a subring of R.

Proof: We have $\varphi(0) + \varphi(0) = \varphi(0+0) = \varphi(0) \Rightarrow \varphi(0) = 0$. If $s_1, s_2 \in \varphi(R)$, then $s_1 = \varphi(r_1), s_2 = \varphi(r_2)$ for $r_1, r_2 \in R$. Then s_1 has inverse $-s_1 = -\varphi(r_1) = \varphi(-r_1) \in \varphi(R)$ and $s_1 + s_2 = \varphi(r_1) + \varphi(r_2) = \varphi(r_1 + r_2) \in \varphi(R)$. Thus $\varphi(R)$ is a subgroup of R. Then $s_1 s_2 = \varphi(r_1)\varphi(r_2) = \varphi(r_1 r_2) \in \varphi(R)$, so $\varphi(R)$ is closed under multiplication. Thus $\varphi(R)$ is a subring of S.

If $a, b \in \ker \varphi$, then $\varphi(-a) = -\varphi(a) = 0$, so $a \in \ker \varphi$, and $\varphi(a+b) = \varphi(a) + \varphi(b) = 0$, so $a + b \in \ker \varphi$. Thus $\ker \varphi$ is a subgroup of R. Now for any $r \in R$, we have $\varphi(ra) = \varphi(r)\varphi(a) = 0 = \varphi(a)\varphi(r) = \varphi(ar)$. Thus $\ker \varphi$ is closed under multiplication and thus a subring of R, and $ra, ar \in \ker \varphi$.

Suppose $T \subseteq S$ is a subring. If $a, b \in \varphi^{-1}(T)$, then $\varphi(a+b) = \varphi(a) + \varphi(b) \in T \Rightarrow a + b \in \varphi^{-1}(T)$ and $\varphi(ab) = \varphi(a)\varphi(b) \in T \Rightarrow ab \in \varphi^{-1}(T)$. It also clearly contains inverses, so it's a subring. ■

Similar to groups, the cosets of the kernel of ring allow us to define the notion of a quotient ring, and there is a simple property we can check to see if a subset of a ring is the kernel of some homomorphism.

Definition (ideal): Let R be a ring, and suppose $I \subseteq R$. For $r \in I$, let

$$rI = \{ra : a \in I\}$$

and

$$Ir = \{ar : a \in I\}.$$

Then I is a *left-sided ideal* if I is a subring of R and $rI \subseteq I$ for all $r \in R$. Similarly, I is a *right-sided ideal* if I is a subring fo R and $Ir \subseteq I$ for all $r \in R$. If I is both a left and right ideal (we often say *two-sided*), then I is an *ideal of* R. If $I \neq R$, then it is proper.

Definition (cosets): For $r \in R$ and an ideal $I \subseteq R$, the set $r + I = \{r + a : a \in R\}$ is a *coset* of I.

Proposition: If $I \subseteq R$ is an ideal and $r_1, r_2 \in I$, then $r_1 + I = r_2 + I$ if and only if $r_1 - r_2 \in I$.

Proof: Suppose $r_1 - r_2 \in I$. If $a \in R$, there exists $b \in I$ such that $a = r_1 + b$. Thus $a - r_1 \in I$. Since I is closed under addition the element $a - r_1 + (r_1 - r_2) = a - r_2$ is in I. Thus $a = r_2 + (a - r_2) \in a_2 + I$, which means $r_1 + I \subseteq r_2 \subseteq I$. The other containment follows similarly. Now suppose $r_1 + I = r_2 + I$. Thus $r_1 + a = c = r_2 + b$ for $a, b \in I$ and $c \in r_1 + I$. Then $r_1 - r_2 = b - a \in I$, as desired. ■

Definition (quotient ring): Given an ideal $I \subseteq R$, define the *quotient ring*

$$R/I = \{r + I : r \in \mathbb{R}\}$$

where $(r_1 + I) + (r_2 + I) = (r_1 + r_2) + I$ and $(r_1 + I) \times (r_2 + I) = r_1 r_2 + I$.

We have to check that the choice of representatives in the definition doesn't matter. Suppose $r_1 + I = s_1 + I$ and $r_2 + I = s_2 + I$. Then $r_1 - s_1, r_2 - s_2 \in I$, so we have

$$(r_1 + r_2 + I) = (r_1 + (s_1 - r_1) + r_2 + (s_2 - r_2) + I) = (s_1 + s_2 + I),$$

so addition is well-defined. For multiplication, since $s_1 - r_1, s_2 - r_2 \in I$, by the definition of an ideal, we see that $r_2(s_1 - r_1), s_1(s_2 - r_2) \in I$. Thus

$$r_1 r_2 + I = r_1 r_2 + r_2(s_1 - r_1) + s_1(s_2 - r_2) + I = s_1 s_2 + I,$$

so multiplication is well-defined as well. Since $0 + I = I$ is the additive identity and $1 + I$ is the multiplicative identity, this is indeed a ring.

Proposition: Let R be a ring and $I \subseteq R$. Then I is the kernel of some homomorphism if and only if it's an ideal.

Proof: We already showed that $\ker \varphi$ was an ideal, so suppose I is an ideal. Then the map $\varphi : R \to R/I$ given by $\varphi(r) = r + I$ has kernel I. ■

Theorem (first ring isomorphism theorem): If $\varphi : R \to S$ is a ring homomorphism, then $R/\ker \varphi \cong \varphi(R)$.

Proof: Let $I = \ker \varphi$, and define $\psi : R/I \to \varphi(R)$ by $\psi(r + I) = \varphi(r)$. Note the choice of representative doesn't matter since $\varphi(r + a) = \varphi(r)$ for $a \in I$. It's easy to check this is a homomorphism. Note that ψ is injective, since $\varphi(r) = 0 \Rightarrow r \in \ker \varphi \Rightarrow \ker \psi = \{0 + I\}$. It's also surjective, since φ outputs all of $\varphi(R)$. Thus the two rings are isomorphic. ■

When working in R/I, we often write $x \equiv y \pmod{I}$ if $x - y \in I$.

Theorem (third ring isomorphism theorem): Let I, J be ideals of R with $I \subseteq J$. Then J/I is an ideal of R/I and $R/J \cong (R/I)/(J/I)$.

Proof: Suppose $j_1 + I, j_2 + I \in J/I$. Since $I \subseteq J$, any representative of these cosets is contained in J. Thus $j_1, j_2 \in J \Rightarrow j_1 + j_2 \in J \Rightarrow (j_1 + j_2) + I \in J/I$. If $r + I \in r/I$, then $(r + I)(j_1 + I) = rj_1 + I \in J/I$ since $rj_1 \in J$. Thus J/I is an ideal in R/I.

Now define $\varphi : R/I \to R/J$ by $\varphi(r + I) = r + J$. Note that this definition is consitent, since if $r_1 - r_2 \in I$, then the difference is also contained in J, so $r_1 + J = r_2 + J$. Clearly φ is surjective. Note that $r + J = 0$ if $r \in J$. Thus $\ker \varphi \subseteq J/I$. Since clearly for $j + I \in J/I$ we have $\varphi(j + I) = j + J = 0$, we see that $\ker \varphi = J/I$. Thus by the first isomorphism theorem, we have

$$(R/I)/(J/I) \cong R/J,$$

as desired. ■

Theorem (fourth ring isomorphism theorem): Let $\varphi : R \to S$ be a surjective ring homomorphism with kernel K. There is a bijective correspondence between the set of all ideals of S and the set of ideals of R that contain K: If I is an ideal of R containing K, then it corresponds to the ideal $\varphi(I)$ in S. If I is an ideal of S, then it corresponds to the ideal $\varphi^{-1}(I)$ in K. If ideals I, J correspond to each other, then $R/I \cong S/J$.

Proof: First we show $\varphi(I), \varphi^{-1}(I)$ are ideals. From the fourth isomorphism theorem for groups, we know that $\varphi(I)$ is additively closed. Pick $y \in \varphi(I)$. Then $\varphi(x) = y$ for some $x \in R$. For

multiplication, we know by surjectivity that for every $s \in S$, there exists $r \in R$ such that $\varphi(r) = s$. Thus $sy = \varphi(r)\varphi(x) = \varphi(rx) \in \varphi(I)$. Thus $\varphi(I)$ is an ideal.

Again by the fourth isomorphism theorem for groups, we know that $\varphi^{-1}(I)$ is additively closed. Pick $r \in R$ and $x \in \varphi^{-1}(I)$. There exists $s, y \in S$ such that $\varphi(r) = s, \varphi(x) = y$. Then $\varphi(rx) = \varphi(r)\varphi(x) = sy$. Thus $rx \in \varphi^{-1}(I)$, so it's an ideal.

The correspondence follows from the fourth isomorphism theorems applied to the additive group.

Now suppose $I \subseteq R, J \subseteq S$ such that $\varphi(I) = J$. There exists the canonical map $\pi : S \to S/J$ given by $\pi(s) = s + J$. Thus $\pi \circ \varphi$ is a map from R to S/J, and is surjective since φ and π are. The kernel consists of elements $r \in R$ such that $\pi(\varphi(r)) = 0 + J \iff \varphi(r) \in J \iff r \in I$. Thus by the first isomorphism theorem, $R/I \cong S/J$. ■

Corollary: Let R be a ring and $a, b \in R$. Then $R/(a, b) \cong (R/(a))/(b)$.

Proof: Let $\varphi : R \to R/(a)$ be a surjective homomorphism (which exists by the first isomorphism theorem). Note that $(a) \subseteq (a, b)$, so by the fourth isomorphism theorem, $\varphi((a, b))$ is an ideal in $R/(a)$. Since $ar + bs \mapsto bs$ under φ, we see it maps to (b) in $R/(a)$. Again by the fourth isomorphism theorem, this implies $R/(a, b) \cong (R/(a))/(b)$. ■

2.1.2. Properties of Ideals

Definition (sum/product of ideals): Let I, J be ideals of R. The *sum of I and J* is

$$I + J := \{a + b : a \in I, b \in J\}.$$

The *product of I and J* is

$$IJ = \left\{a_1 b_1 + \cdots + a_n b_n : a_i \in I, b_j \in J\right\}.$$

It's easy to check these are ideals. Note that if I, J are finitely generated, then we can check that IJ is generated by the pairwise products of the generators.

Definition (generated ideals): If $A \subseteq R$, then let the smallest ideal of R containing A, called the *ideal generated by A*, be denoted by (A). If $A = \{a_1, ..., a_n\}$, we often write $(a_1, ..., a_n)$. If A is finite, then (A) is said to be *finitely generated*.

Proposition: Suppose $A = \{a_1, ..., a_n\} \subseteq R$. Then

$$(a_1, ..., a_n) = \{r_1 a_1 + \cdots + r_n a_n : r_i \in R\}.$$

Proof: Clearly the right side is an ideal that contains A, so $(a_1, ..., a_n) \subseteq \{r_1 a_1 + \cdots + r_n a_n : r_i \in R\}$. Since clearly every element of the right is an element of the left (since the left is an ideal), we see the two sets are equal. ■

Proposition: Let I be an ideal of R. Then I is proper if and only if it contains no units. Furthermore, if R is commutative, then R is a field if and only if its only ideals are R and 0.

Proof: If I isn't proper, it clearly contains units. Suppose $u \in I$ is a unit. Then $1 = uu^{-1} \in I$, so $r \cdot 1 \in I$ for all $r \in R$.

Now suppose R is a field. Since every nontrivial ideal must contain a unit, by the first claim the ideal is R, so those are the only two ideals. Conversely, suppose the only ideals are 0 and R, and pick nonzero $u \in R$. Then $(u) = R$, so $1 \in (u)$. Thus there exists $v \in R$ such that $vu = 1$, so u is a unit. Since u was arbitrary, we see that R is a field. ■

Corollary: If R is a field, then any nonzero ring homomorphism from R into another ring is injective.

Proof: Since the kernel of a homomorphism is an ideal, and since R only has two ideals, the kernel is either R or 0. Since the homomorphism is nontrivial, it must be 0. ■

Definition (principal/maximal/prime ideal): Let $I \subseteq R$ be an ideal.

- If $I = (a)$ for some $a \in R$, then I is a *principal ideal.*
- If I is proper and there exists no other proper ideal containing I, then I is a *maximal ideal.*
- Suppose R is commutative and I is proper. If $xy \in I$ implies either $x \in I$ or $y \in I$ for all $x, y \in R$, then I is a *prime ideal.*

Proposition (every ring has a maximal ideal): Every ring R contains a maximal ideal.

Proof: To show this, we need Zorn's lemma. Partially order the proper ideals of R by inclusion. Suppose we have some chain $\mathcal{C}$ of ideals in R. Let $J = \bigcup_{I \in \mathcal{C}} I$. We show that J is an ideal. Indeed, if $a, b \in J$, then since $\mathcal{C}$ is a chain, there must exists $I \in \mathcal{C}$ for which $a, b \in I$, which implies $a + b \in I$, which implies $a + b \in J$. Since J clearly cotains 0, its an additive subgroup. Similarly if $a \in J$, then $a \in I$ for some $I \in \mathcal{C}$, so $ra, ar \in I$ for all $r \in R$, so $ra, ar \in J$ for all $r \in R$, so J absorbs multiplication from both sides. Thus J is an ideal, and it can't contain 1, since otherwise some element of $\mathcal{C}$ would contain 1 and thus be nonproper, and so J is also proper, and thus is an upper bound for $\mathcal{C}$.

Since every chain has an upper bound. Thus by Zorn's lemma, there exists some maximal element M that's an ideal of R. Since this means that no proper ideal contains M, it must be a maximal ideal, as desired. ■

Proposition: If R if commutative, then $M \subseteq R$ is a maximal ideal if and only if R/M is a field.

Proof: The fourth isomorphism theorem implies every ideal of R/M corresponds to an ideal of R containing M. Since M is maximal, the only ideals that do so are M and R. These map to (0) and R/M in R/M. Thus it only has two ideals, and so is a field. If R/M is a field, doing this in reverse yields that M is maximal. ■

Proposition: If R is commutative and I is a proper ideal of R, the I is a prime ideal if and only if R/I is an integral domain.

Proof: Suppose I is prime, and suppose $ab \equiv 0 \pmod{I}$ for $a, b \in R/I$. Thus $ab \in I$, which implies either a or b is in I. Without loss of generality, suppose $a \in I$. Then $a \equiv 0 \pmod{I}$. Thus R/I has no zero divisors, so it's an integral domain.

Now suppose R/I is an integral domain, and suppose $a, b \in R$ such that $ab \in I$. Then $ab \equiv 0 \pmod{I}$, and since R/I is an integral domain, this implies one of a, b is $0 \pmod{I}$. This implies one of a, b is in I, so I is a prime ideal. ■

Proposition: If R is commutative, then every maximal ideal of R is a prime ideal.

Proof: If $M \subseteq R$ is maximal, then R/M is a field, and thus an integral domain, and thus M is prime. ■

2.1.3. Chinese Remainder Theorem

Definition (product ring): If R, S are rings, then $R \times S$ is a ring with addition and multiplication performed pointwise.

Definition (comaximal): The ideals $A, B \subseteq R$ are *comaximal* if $A + B = R$.

Proposition: Let $A_1, ..., A_k \subseteq R$ be ideals in a commutative ring R. Then the map $\varphi : R \to R/A_1 \times \cdots \times R/A_k$ defined by

$$\varphi(r) = (r + A_1, ..., r + A_k)$$

is a ring homomorphism and has kernel $A_1 \cap \cdots \cap A_k$.

Proof: Checking that φ is a homomorphism is easy, and note that $\varphi(r) = 0$ if and only if $r \in A_1, A_2, ...A_k \Rightarrow r \in A_1 \cap \cdots \cap A_k$. ■

Theorem (Chinese remainder theorem): Suppose the ideals $A_1, \ldots, A_k \subseteq R$ in a commutative ring are pairwise comaximal. Then $A_1 \cap \cdots \cap A_k = A_1 A_2 \cdots A_k$ and

$$R/(A_1 A_2 \cdots A_k) \cong R/A_1 \times R/A_2 \times \cdots \times R/A_k.$$

Proof: We prove the case when $k = 2$, as the general case follows by induction. Let $I, J \subseteq R$ be comaximal ideals. Thus there exists $i \in I, j \in J$ such that $i + j = 1$. Thus if $r \in I \cap J$, then $ri \in I, rj \in J$, so $r = r(i + j) = ri + rj \in IJ$. Thus $I \cap J \subseteq IJ$. If $r \in IJ$, there exists $a_i \in I, b_i \in J$ such that $a_1 b_1 + \cdots + a_n b_n = r$, and from the definition of an ideal, we see that $r \in I, J$. Thus $I \cap J = IJ$.

Now consider $\varphi : R/(IJ) \to R/I \times R/J$ defined in the previous proposition. Note that for any $a, b \in R$, we have $\varphi(ai + bj) = (ai + bj + I, ai + bj + J) = (b(1 - i) + I, a(1 - j) + J) = (b + I, a + J)$, so φ is surjective. Then by the first isomorphism theorem,

$$R/(IJ) = R/(I \cap J) \cong R/I \times R/J.$$

■

2.2. Flavors of Rings

2.2.1. Euclidean Domains

Definition (norm): For integral domains R, a function $N : R \to \mathbb{Z}_{\geq 0}$ is called a *norm* if $N(0) = 0$. If $N(a) > 0$ for $a \neq 0$, then it's called a *positive norm*.

Definition (Euclidean domain): An integral domain R is a *Euclidean domain* if there exists a norm on R such that for any two elements $a, b \in R$ with $b \neq 0$, there exists $q, r \in R$ such that

$$a = qb + r \quad \text{and} \quad r = 0 \text{ or } N(r) < N(b).$$

Example: $\mathbb{Z}$ is the best example of a Euclidean domain, with norm being the absolute value. Division occurs via the Euclidean algorithm. Another important example is $F[x]$, where F is a field. This time the norm is the degree of the polynomial. We'll prove later that this norm actually makes $F[x]$ a Euclidean domain.

Proposition: Every ideal in a Euclidean domain is principal.

Proof: If $I \subseteq R$ is the zero ideal in a Euclidean domain, then it's obviously principal. Suppose I is nonzero, and let d be a nonzero element of minimal norm in I (which must exist by well-ordering). We claim that $(d) = I$. Clearly the left is a subset of the right. Now suppose $a \in I$.

Since R is Euclidean, there exists $q, r \in R$ for which $a = qd + r$ with $N(r) < N(b)$. Since $a, d \in I$, we must have $a - qd = r \in I$. Since $N(r) < N(d)$, we must have that $r = 0$. Thus $a = qd \in (d)$, so $I \subseteq (d)$, as desired. ■

Example: Consider $\mathbb{Z}[x]$ and the ideal $(2, x)$. If this were principal, say (p) for some $p \in \mathbb{Z}[x]$, then it would have to have degree zero, since $pq = 2$ for some $q \in \mathbb{Z}[x]$. Thus $(p) = (n)$ for some integer n. It can't be 1 since $(1) = \mathbb{Z}[x] \neq (2, x)$ (as the right doesn't contain 3). It obivously can't be an integer > 2, so n must be 2. But then any polynomial in (2) will have even coefficients, so it won't contain x. Thus $(2, x)$ isn't principal, so $\mathbb{Z}[x]$ can't be a Euclidean domain.

Example: Consider $\mathbb{Z}\left[\sqrt{-5}\right]$, and give it norm $N\left(a + b\sqrt{-5}\right) = a^2 + 5b^2$ (this norm comes from the norm on $\mathbb{C}$). We can check that this norm is multiplicative. Using this norm, we'll show that the ideal $I = \left(3, 2 + \sqrt{-5}\right)$ is not principcal, which then implies that $\mathbb{Z}\left[\sqrt{-5}\right]$ is not a Euclidean domain.

Suppose $I = \left(a + b\sqrt{-5}\right)$. Thus there exists $\alpha, \beta \in \mathbb{Z}\left[\sqrt{-5}\right]$ such that $3 = \alpha\left(a + b\sqrt{-5}\right)$ and $2 + \sqrt{-5} = \beta\left(a + b\sqrt{-5}\right)$. Taking the norm of the first equation yields $9 = N(\alpha)(a^2 + 5b^2)$. Since the norms are positive integers, we see that $a^2 + 5b^2 = 1, 3, 9$. This has no solutions for $a^2 + 5b^2 = 3$, and if $a^2 + 5b^2 = 1$, then $a + b\sqrt{-5} = \pm 1$, which would imply $I = \mathbb{Z}\left[\sqrt{-5}\right]$. But we can check that I isn't all of $\mathbb{Z}\left[\sqrt{-5}\right]$ (in particular it doesn't contain 1).

If $a^2 + 5b^2 = 9$, then $N(\alpha) = 1$, which implies $\alpha = \pm 1$. Thus $a + b\sqrt{-5} = \pm 3$. However, this can't work since by the second equation, this would implies $2 + \sqrt{-5} = \pm 3\left(a + b\sqrt{-5}\right)$, which is impossible since 3 doesn't divide the coefficients of the left side. Thus I is not principal.

Proposition: If F is a field, then $F[x]$ is a Euclidean domain.

Proof: Let $N : F[x] \backslash \{0\} \to \mathbb{Z}_{\geq 0}$ by $N(p(x)) = \deg p$. Suppose $p, q \in F[x]$ with $q \neq 0$. If $\deg p < \deg q$, then $p = 0 \cdot q + p$, so this satisied the condition. If $\deg q = 0$, then q si a constant, so $q \mid p$. Otherwise suppose $\deg q \geq 1$, and suppose $\deg p \geq \deg q$, and let $p(x) = a_n x^n + \cdots + a_0$ and $q(x) = b_m x^m + \cdots + b_0$, where $n = \deg p, m = \deg q$. Let $s(x) = \frac{a_n}{b_m} x^{n-m}$. Then $r = p - qs$ has degree at most $n - 1$. If $\deg r < \deg q$, then we're done. Otherwise suppose $r(x) = c_{\deg r} x^{\deg r} + \cdots + c_0$. Let $t(x) = c_{\deg r}/b_m x^{\deg r - m}$. Thus $u = r - qt = p - q(t - s)$ has degree at most $\deg r - 1$. Repeating thing, we see that the degree decreases every step, so eventually the degree will be less than m, and the condition will be satisfied. ■

Definition (ring divisibility): Let R be a commutative ring and let $a, b \in R$ with $b \neq 0$. a is said to be a *multiple* of b if there exists an element $x \in R$ with $a = bx$, and b is said to *divide* a, denoted $b \mid a$.

Definition (ring gcd): Let R be a commutative ring and let $a, b \in R$. A *greatest common divisor* of a and b is a nonzero element d such that $d \mid a, b$ and such that if $d' \mid a, b$, then $d' \mid d$. If it exists, the gcd is denoted $\gcd(a, b)$.

Note that from the definition of divisibility, we see that $b \mid a$ if and only if $(a) \subseteq (b)$ (which feels wonky but it the correct inclusion). Using this, we can also give an equivalent definition of the gcd. If $I = (a, b)$ then d is the gcd of a, b if $I \subseteq (d)$ and if $I \subseteq (d')$, then $(d) \subseteq (d')$.

We've been referring to *the* gcd, and it is essentially unique in integral domains.

Proposition: Let R be an integral domain. If two elements $d, d' \in R$ generate the same ideal, i.e. $(d) = (d')$, then $d' = ud$ for some unit $u \in R$.

In particular, if $d, d' = \gcd(a, b)$, then $d' = du$ for some unit $u \in R$.

Proof: Obvious for $d, d' = 0$, so assume they're nonzero. Thus there exists $x, y \in R$ such that $d' = dx$ and $d = d'y$. Thus $d' = d'xy$. Since R is an integral domain and d' is nonzero, we obtain $xy = 1$. Thus x, y are units, so we're done. ■

Using this, we can basically do the Euclidean algorithm in Euclidean domains (using the norm).

Definition (universal side divisor): Let R be an integral domain. An element $u \in R \backslash (R^\times \cup \{0\})$ is called a *universal side divisor* if for every $x \in R$, there exissts $z \in R^\times \cup \{0\}$ $u \mid x - z$. In other words, x can be written as $x = qu + z$, where z is zero or a unit.

Proposition: Let R be an integral domain that's not a field. If R is a Euclidean domain, there exists a universal side divisor in R.

Proof: Let N be a Euclidean norm for R, and let u be an element of minimal norm that's not zero or a unit. For any $x \in R$, we can write $x = qu + r$ with $N(r) < N(u)$. Since u has minimal nonzero and non-unit norm, this implies $r \in R^\times \cup \{0\}$, so u is a universal side divisor, as desired. ■

Example: We show that $\mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$ is not Euclidean (it turns out to have only principal ideals, though) buy showing it contains no universal side divisors. Let $N\left(a + b \cdot \frac{1+\sqrt{19}}{2}\right) = a^2 + ab + 5b^2 = \left(a + \frac{b}{2}\right)^2 + \frac{19}{4}b^2$, which is its modulus in $\mathbb{C}$. Thus it is multiplicative. If r is a unit, then $1 = N(rr^{-1}) = N(r)N(r^{-1})$. Thus units have norm ± 1. Similary, if $N(r) = \pm 1$, we see that $a = \pm 1, b = 0$. Thus r is a unit in $\mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$ if and only if $r = \pm 1$. We also see that $N(r) = 0$ if and only if $r = 0$.

We see that the smallest nonzero, non-one norm value is 4, given by ± 2. Thus if there exists a universal side divisor u it would either divide 2 or $2 \pm 1 = 1, 3$. Writing $2 = ab$, we see that $4 = N(a)N(b)$, and by the minimality above, this implies that either $N(a)$ or $N(b)$ is ± 1, and thus either a or b is ± 1. Thus the other number is ± 2. Similarly, the only non unit divisors of 3 are ± 3. Thus a universal side divisor must be $\pm 2, \pm 3$. We can check that for $x = \frac{1+\sqrt{-19}}{2}$, none of these divide $x, x \pm 1$. Thus there are no universal side divisors, so $\mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$ is not a Euclidean domain.

2.2.2. Principal Ideal Domains

Definition (principal ideal domain): A *principal ideal domain* (or PID) is an integral domain in which every ideal is principal.

Thus earlier we proved every Euclidean domain is a PID.

Example: We showed that $\left(3, 1+\sqrt{-5}\right)$ is not principal in $\mathbb{Z}\left[\sqrt{-5}\right]$. Similarly we can show that $\left(3, 1-\sqrt{-5}\right)$ is not principal. However, the product is

$$\left(3, 1+\sqrt{-5}\right)\left(3, 1-\sqrt{-5}\right) = \left(9, 3+3\sqrt{-5}, 3-3\sqrt{-5}, 6\right).$$

From the definition of generated ideals, we can subtract an multiple from one generator of an element to another. Thus the ideal is equal to

$$\left(3, 3+3\sqrt{-5}, 3-3\sqrt{-5}, 6\right) = (3),$$

since each generator here is a multiple of 3. Thus the product of two non-principal ideals can be principal.

Proposition (prime ideals are maximal in PIDs): Every nonzero prime ideal in a PID is maximal.

Proof: Let $I \subseteq R$ be a nonzero prime ideal, and suppose there exists an ideal J such that $I \subseteq J \subseteq R$. Since R is a PID, we can write $I = (a)$, $J = (b)$ for some $a, b \in R$. Thus $a = bx$ for some $x \in R$. Thus $bx \in I$, and since I is a prime ideal, either $b \in I$ or $x \in I$. In the first case, this implies $(b) \subseteq (a)$, so $(b) = (a)$. In the other case, there exists $r \in R$ such that $x = ar$. Thus $a = bra \Rightarrow br = 1$. Thus b is a unit, so $J = R$. Thus there exists no propert ideal strictly containing I, so it's maximal. ■

Corollary: For commutative ring R, if $R[x]$ is a PID, then R is a field.

Proof: Suppose $R[x]$ is a PID. Since R is a subring of $R[x]$, it must also be an integral domain. Suppose $pq \in (x)$ for $p, q \in R[x]$. Then pq has no constant term, so at least one of p, q can't either. Thus one of them is in (x), so (x) is a prime ideal, and thus maximal since $R[x]$ is a PID. This implies $R[x]/(x)$ is a field. The map $\varphi : R \to R[x]/(x)$ given by $\varphi(r) = r + (x)$ is easily seen to be a bijective homomorphism, so the two are isomorphic. Thus R is a field. ■

2.2.3. Unique Factorization Domains

Definition (irreducible): Let R be an integral domain. Suppose $r \in R$ is nonzero and not a unit. Then r is *irreducible* in R if whenever $r = ab$ with $a, b \in R$, at least one of a, b is a unit in R. Otherwise r is *reducible*.

Definition (prime element): Let R be an integral domain. An element $p \in R$ is *prime* in R if the ideal (p) is a prime ideal.

Using the definition of a prime ideal, we see that p is prime if it's not a unit and if $p \mid ab$ for $a, b \in R$, then $p \mid a$ or $p \mid b$.

Definition (associate): Two elements a, b of a ring R are said to be *associate* in R if $a = ub$ for some unit $u \in R$.

Proposition: In an integral domain R, every prime element is irreducible.

Proof: Suppose $p \in R$ is a prime element and $p = ab$. Then Clearly $p \mid ab$, so one of a, b is divisible by p, without loss of generality say a. Then $a = rp$ for some $r \in R$. Thus $p = rbp \Rightarrow rb = 1$, so b is a unit. Thus p is irreducible. ■

Example *(not every irreducible is prime)*: Consider $3 \in \mathbb{Z}\left[\sqrt{-5}\right]$. Suppose $3 = ab$. Using the norm on $\mathbb{Z}\left[\sqrt{-5}\right]$, we obtain $9 = N(a)N(b)$. Note that $c^2 + 5d^2 = 3$ has no integer solutions, so the norm can't be 3. Thus one of $N(a), N(b)$ must be 1, which implies one of them is a unit, so 3 is irreducible. However, $\left(2 + \sqrt{-5}\right)\left(2 - \sqrt{-5}\right) = 3^2$ is divisible by 3, but neither factor is.

Proposition: If R is a PID, then a nonzero element is prime if and only if it's irreducible.

Proof: We already showed prime implies irreducible, so suppose $r \in R$ is irreducible. If M is any ideal containing (r), then $M = (a)$ for some $a \in R$, so $r = am$ for some $m \in R$. Since r is irreducible, this implies either a or m is a unit. The first case implies $M = R$, the second case implies $M = (a)$. Thus (r) is maximal, and thus prime, so r is a prime element. ■

Definition (unique factorization domain): A *unique factorization domain* (or UFD) is an integral domain R in which every nonzero not unit element $r \in R$ has the following two properties:

a) r can be written as a finite product of not necessarily irreducibles p_i of R.
b) The decomposition of r is unique up to associates. That is, if $r = p_1 p_2 \ldots p_n = q_1 q_2 \ldots q_m$ are factorizations into irreducibles, then $m = n$ and there is a relabeling of factors such that p_i and q_i are associate.

Proposition: Let R be a UFD. Then every nonzero element is prime if and only if it's irreducible.

Proof: We already know primes are irreducible, so let $p \in R$ is irreducible, and suppose $p \mid ab$ for $a, b \in R$. Thus $pc = ab$ for some $c \in R$. Since we're in a UFD, we can write a and b as the

product of irreducibles. Then from the uniqueness of factorization, and since p is irreducible, we see that p is associate to some irreducible in the factorization of ab, and thus associate to some irreducible in the factorization of a or b, and thus divides a or b (the unit can be ignored since it ends up in the dividend). Thus p is prime. ■

Proposition (PID $\Rightarrow$ UFD): Every principal ideal domain is a unique factorization domain.

Proof: Let R be a PID and let $r \in R$ be nonzero and not a unit. We first show that r can be written as a finite product of irreducibles up to units, and then show the decomposition is unique up to units.

If r is irreducible, then we're done. Otherwise, we can write $r = r_1 r_2$ where neither factors are units. If both of these are irreducible, we're done. Otherwise, at least one of them, say r_1 is reducible, and we can write $r_1 = r_{11} r_{12}$, where the factors are not units. We can continue this process. Suppose for the sake of contradiction this process doesn't terminate. From the factorization $r = r_1 r_2$, we get the inclusions $(r) \subseteq (r_1) \subseteq R$. Note that since r_2 is not a unit, the first inclusion is proper. Similarly, since r is not a unit, the last inclusion is proper. Doing the same with r_1 then yields the containments $(r) \subset (r_1) \subset (r_{11}) \subset R$. Since the process doesn't terminate, we get an infinite chain of strict containments

$$(r) \subset (r_1) \subset (r_{11}) \subset \cdots \subset R.$$

We claim that an ascending chain of ideals $I_1 \subseteq I_2 \subseteq \cdots \subseteq R$ must eventually become constant in a PID, thus contradicting the above chain and proving the process of factorizing r terminates. Let $I = \bigcup_{n=1}^{\infty}$. Then I is an ideal, so $I = (a)$. Thus $a \in I_k$ for some k. Thus $(a) \subseteq I_k \subseteq I = (a)$, so $I = I_k$, and thus $I_n = I$ for all $n \geq k$, as desired.

Now we show the decomposition is unique up to units. We induct in the number of irreducibles in the facotization. If $n = 0$, then r is a unit. If were to be $r = qc$ for some irreducible q, then q would divide a unit, thus making it a unit, contradiction. Thus the factorization is unique, so the base case is done. Now suppose $n \geq 1$ and

$$r = p_1 p_2 \cdots p_n = q_1 q_2 \cdots q_n$$

with $m \geq n$, where p_i, q_j are irreducibles. Since we're in PID, p_1 is a prime element, so it must divide one of the q_j, without loss of generality say q_1. Thus $q_1 = p_1 u$, and since q_1 is irreducible and p_1 is not a unit, u must be a unit. Thus p_1, q_1 are associates. Cancelling p_1 yields

$$p_2 \cdots p_n = u q_2 \cdots q_m.$$

Letting $q_2' = u q_2$, we have a factorization of some element into $n-1$ irreducibles. Applying the induction hypotheses yields that $m - 1 = n - 1 \Rightarrow m = n$ and that the p_i and q_j or q_2' are associate via some bijection of elements. Since u is a unit, this means the p_i and q_i are associates for $i, j \geq 2$, and since p_1 and q_1 are associates, this holds for $i, j \geq 1$. Thus the decomposition is unique, as desired. ■

Definition (Dedekind-Hasse norm): Define N to be a *Dedekind-Hasse norm* if N is a positive norm and for every nonzero $a, b \in R$, either $a \in (b)$ or there is a nonzero element in the ideal (a, b) of norm strictly smaller than $N(b)$.

In other words, either $b \mid a$ or there exists $s, t \in R$ such that $0 < N(sa - tb) < N(b)$.

Proposition: An integral domain R is a PID if and only if R has a Dedekind-Hasse norm.

Proof: Suppose R admits a Dedekind-Hasse norm N. Let I be a nonzero ideal in R and let b be a nonzero element of I with minimal norm. Pick nonzero $a \in I$. Thus $(a, b) \subseteq I$, which means there can't exist a nonzero element in (a, b) with norm less than b. Thus by the definition of a Dedekind-Hasse nrom, we must have $a \in (b)$. Since this holds for all $a \in I$, we see that $I = (b)$, so I is principal. Thus I is a PID.

Now suppose R is a PID. Then from above, R is a UFD. Define a norm on R by setting $N(0) = 0$, $N(u) = 1$ if $u \in R^\times$, and $N(a) = 2^n$ if $a = p_1 \cdots p_n$ for p_i irreducible. This is well-defined since the number of irreducible factors is unique by definition of a UFD. Clearly N is multiplicative and positive. Let $a, b \in R$ be nonzero. Then $(a, b) = (r)$ for some $r \in R$. If $a \notin (b)$, then (r) strictly contains (b), so r is also not in (b). Since $b = xr$ for some $x \in R$, the previous statement implies that x is not a unit. Thus $N(b) = N(x)N(r) > N(r)$, so (a, b) contains a nonzero element with norm strictly less than $N(b)$. Thus N is a Dedekind-Hasse norm. ■

2.2.4. Field of Fractions

Definition (field of fractions): Let R be an integral domain. The *field of fractions* of R, denoted $\mathrm{Frac}(R)$, is defined as follows: it consists of elements a/b with $a, b \in R$ and $b \neq 0$. We set $a/b \sim c/d$ if and only if $ad = bc$. Addition and multiplication are defined as

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

Since these fields basically behave like $\mathbb{Q}$, it's easy to check that the definition does indeed create a field.

Example: Consider $\mathrm{Frac}(\mathbb{Z}[i])$. This consists of elements $\frac{a+bi}{c+di}$. By rationalizing the denominator, we get an element of $\mathbb{Q}[i]$. Indeed, $\mathrm{Frac}(\mathbb{Z}[i]) = \mathbb{Q}(i)$.

2.2.5. Polynomial Rings

Proposition: Let R be an integral domain. Then $R[x]^\times = R^\times$ and $R[x]$ an an integral domain.

Proof: Clearly $R^x \subseteq R[x]^\times$. If $p(x)q(x) = 1$, then both have to be degree 0, so both are constants, and thus both are in R. Thus $R[x]^\times \subseteq R^\times$. If $p(x)q(x) = 0$ and are nonzero, then both can't have degree greater than 0, so they both must be constants. Since R is an integral domain, this implies one of them is 0, so $R[x]$ is an integral domain. ■

Proposition: Let I be an ideal of R, and let $(I) = I[x]$ be the ideal of $R[x]$ generated by I. Then $R[x]/(I) \cong (R/I)[x]$. Furthermore, if I is a prime ideal of R, then (I) is a prime ideal of $R[x]$.

Proof: Consider the map $\varphi : R[x] \to (R/I)[x]$, where we reduce the coefficients mod I. This is clearly a surjective ring homomorphism with kernel (I). Thus by the first isomorphism theorem, $R[x]/(I) \cong (R/I)[x]$. If I is a prime ideal, then R/I is an integral domain, and thus so is $(R/I)[x]$, and thus so is $R[x]/(I)$, so (I) is also a prime ideal. ■

If F is a field, we already showed that $F[x]$ is a Euclidean domain. Thus from previous results, it's also a PID and UFD.

Lemma (Gauss' Lemma): Let R be a UFD with field of fractions F and let $p(x) \in R[x]$. If $p(x)$ is reducible in $F[x]$, then $p(x)$ is reducible in $R[x]$.

Proof: Suppose $p(x) = A(x)B(x)$ for nonconstant $A, B \in F[x]$. Since we're in a UFD, we can take the gcd of all denominators of coefficients of A and B, say d. Multiplying by d yields $dp(x) = a'(x)b'(x)$, with $a', b' \in R[x]$. If d is a unit, then we're done, so suppose otherwise. We can factor d as $d = p_1 \cdots p_n$. Since p_1 is irreducible, the ideal (p_1) is prime in R, so $R/(p_1)$ is an integral domain. Thus $(R/(p_1))[x]$ is an integral domain. Reducing the equation mod p_1 yields $0 \equiv a'(x)b'(x) \pmod{p_1}$. Since $(R/(p_1))[x]$ is an integral domain, one of the two factors, say $a'(x)$, must be zero mod p_1. Thus all of its coefficients are divisible by p_1. Thus we can divide the equation in $R[x]$ by p_1 and still have an equation in $R[x]$. We can do this for every factor of d, and eventually we'll have $p(x) = a(x)b(x)$ for $a, b \in R[x]$ being F-multiples of A, B, so p is reducible in $R[x]$, as desired. ■

Corollary: Let R be a UFD with field of fractions F, and let $p(x) \in R[x]$. Suppose the gcd of the coefficients of p is 1. Then $p(x)$ is irreducible in $R[x]$ if and only if it's irreducible in $F[x]$. In particular, if $p(x)$ is a monic polynomial that's irreducible in $R[x]$, then $p(x)$ is irreducible in $F[x]$.

Proof: By Gauss' lemma, if $p(x)$ is reducible in $F[x]$, then it's reducible in $R[x]$. Now suppose $p(x)$ is irreducible in $R[x]$. By the gcd assumption, we must have $p(x) = a(x)b(x)$ for a, b being nonconstant. Thus a, b are not units in $F[x]$, so p is reducible in $F[x]$, as desired. ■

Proposition: R is a UFD if and only if $R[x]$ is a UFD.

Proof: It's clear that if $R[x]$ is a UFD, then R is a UFD, so suppose R is a UFD. Let F denote the field of fractions of R and let $p(x) \in R[x]$ be nonzero. Let d be the gcd of the coefficients of p. Thus $p(x) = dp'(x)$. Since $d \in R$, we can uniquely factor it into irreducibles. Thus we may assume that $p(x)$ has coefficients with gcd 1.

If $p(x)$ is a unit, then we have nothing to do, so we can assume $\deg p > 0$. Since $F[x]$ is a UFD, $p(x)$ can be uniquely factored into irreducibles in $F[x]$. By Gauss' lemma, the existence of this

factorization implies that there is a factorization of $p(x)$ in $R[x]$ whose factors are F-multiples of factors in $F[x]$. Since the gcd of the coefficients is 1, the gcd of the coefficients in each of these factors must be 1. Then by the corollary, each of these factors is irreducible in $R[x]$. Thus $p(x)$ can be written as the finite product of irreducibles in $R[x]$.

Now suppose

$$p(x) = q_1(x) \cdots q_r(x) = q_1'(x) \cdots q_s'(x)$$

are factorizations of $p(x)$ in $R[x]$ into irreducibles. Since the gcd of the coefficients of p is 1, the gcd of the coefficients of the factors is 1. Thus each of them has nonzero degree. By the corollary, the factors are thus irreducible in $F[x]$. Thus since $F[x]$ is a UFD, $r = s$ and, after relabeling, q_i and q_i' are associate in $F[x]$ for $i = 1, \ldots, r$. Thus we just need to show the factors are associate in $R[x]$.

Since the units of $F[x]$ is $F^\times$, we have $q_i(x) = \frac{a}{b} q_i'(x)$ for $q_i, q_i' \in R[x]$ with the gcd of the coefficients being 1 and $a, b \in R$ nonzero. Thus $bq_i(x) = aq_i'(x)$. Now the gcd of the coefficients of the left is b and the gcd of the coefficients of the right is a. Since R is an integral domain, the gcd is unique up to units, so $a = ub$ for some unit $u \in R$. Thus $q_{i(x)} = uq_i'(x)$, so q_i and q_i' are associates in $R[x]$, as desired. ■

We spend the rest of this section discussing irreducibility in $F[x]$ (since by Gauss' lemma, irreducibility in $R[x]$ can be checked by factoring in $F[x]$). The rational root theorem ports over from $\mathbb{Q}$ basically using the same proof, so we don't state it here.

Proposition: Let I be a proper ideal in an integral domain R and let $p(x)$ be a nonconstant monic polynomial in $R[x]$. If $p(x)$ with coefficients reduced mod I is not factorable into two polynomials of smaller degree in $(R/I)[x]$, then $p(x)$ is irreducible in $R[x]$.

Proof: Suppose that $p(x)$ can't be factored in $(R/I)[x]$ but is reducible in $R[x]$. Thus we can write $p(x) = a(x)b(x)$ where $a, b \in R[x]$ are nonconstant and monic (we can make them monic since their leading coefficients must be units in R that are inverses of each other). Then reducing the coefficients mod I yields $p(x) \equiv a(x)b(x) \pmod{I}$. Since a, b are monic, these are nonzero mod I, so we have a contradiction. ■

Proposition (Eisenstein's criterion): Let P be a prime ideal of an integral domain R and let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial in $R[x]$ with $n \geq 1$. If $a_{n-1}, \ldots, a_1, a_0$ are elements of P and a_0 is not an element of P^2, then $f(x)$ is irreducible in $R[x]$.

Proof: Suppose that f was reducible with $f(x) = a(x)b(x)$ in $R[x]$. Since f is monic, a, b are monic and thus nonconstant. Reducing mod P yields $x^n \equiv a(x)b(x) \pmod{P}$ in $(R/P)[x]$ with a, b still being nonconstant since they're monic. Thus both have degree less than n, so x divides both of them mod P. Thus both have constant 0 mod P, which means both have constants in P. But this implies that $a_0 \in P^2$, contradiction. ■

2.3. Problems

Problem: If R is a ring with identity, prove that $(-1)^2 = 1$.

Solution: $(-1) \times (-1) = 1 \times 1 = 1$.

Problem: Prove the quotient of a PID of a prime ideal is a PID.

Solution: This follows since prime ideals are maximal in PIDs, so the quotient is a field, which only has trivial ideals.

Problem: Let R be an integral domain and suppose that every prime ideal in R is principal. Prove that R is a PID.

Steps pulled form subparts of Dummit and Foote.

Proof: Let N be the set of ideals in R that are not principal. If it's empty then we're done, so assume otherwise. Partially order N by inclusion, and suppose $\mathcal{C}$ is some chain. Then we know that $\bigcup \mathcal{C}$ is an ideal that contains all elements of $\mathcal{C}$. If $\bigcup \mathcal{C}$ was principal, say equal to (a), then some $I \in N$ would contain a, and so $(a) \subseteq I \subseteq J = (a)$, which implies I is principal, contradiction. Thus J must be not principal, so $J \in N$. Thus every chain in N has a maximum, so by Zorn's lemma, there exists a maximal element $I \in N$.

Note that if I was a prime ideal, it would principal by hypothesis, so there must exist $a, b \in R$ such that $ab \in I$ by $a, b \notin I$. Let $I_a = (I, a)$ be the ideal generated by I and a. Clearly $I \subset I_a$, and since I is a maximal element of N, I_a must be principal, say equal to (α) for $\alpha \in R$. Similarly, $I_b = (I, b)$ is equal to (β).

Now define $J = \{r \in R : rI_a \subseteq I\}$. It's easy to check that J is an ideal. Now suppose $r \in I_b$. Thus we can write it as $i + sb$, where $i \in I, s \in R$. Now for any $i' \in I, t \in R$, we have

$$(i + sb)(i' + ta) = ii' + sbi' + tai + stab.$$

Each term is in I, so the whole sum is in I. Thus $(i + ab)I_a \subseteq I$, so $I_b \subseteq J$. Again since I is a maximal non principal ideal, we see that $J = (\gamma)$ for some $\gamma \in R$. Then $I_a J = (\alpha\gamma) \subseteq I$.

Suppose $x \in I$. Since $I \subseteq I_a$, there exists $r \in R$ such that $x = r\alpha$. Thus $rI_a = r(\alpha) = (r\alpha) = (x) \subseteq I$. Thus $r \in J$, which means $r = s\gamma$ for some $s \in R$. Thus $x = s\alpha\gamma \in (\alpha\gamma)$. Since x was arbitrary, we see that $I \subseteq (a\gamma)$. Thus $I = (a\gamma)$, so I is principal, contradiction. Thus N must be empty, so R is a PID, as desired. ■

Problem: If R is a UFD in which every prime ideal is maximal, show that R is a PID.

Solution: Suppose I is a prime ideal. Since we're in a UFD, there exists some nonzero element $r = p_1 \cdots p_n$ with a minimal number of prime factors. If $n \geq 2$, then $p_1 \cdot (p_2 \cdots p_n) \in I$ but $p_1, p_2 \cdots p_n \notin I$, so I would not be prime. Thus $n = 1$, so $r = p$ for some prime p. Thus $(p) \subseteq I$. Since p is prime, (p) is prime, thus maximal, so $(p) = I$. Thus every prime ideal is principal, so by the previous problem, we see that R is a PID, as desired.

Problem: Let $R = \mathbb{Z}[\sqrt{-n}]$, where $n > 3$ is squarefree.

a) Prove that $2, \sqrt{-n}, 1 + \sqrt{-n}$ are irreducible in R.
b) Prove that R is not a UFD.
c) Give an explicit ideal in R that is not principal.

Solution:

a) Let N be the norm on R given by $a^2 + nb^2$. Since N is multiplicatice, for a unit u, we have $N(u)N(u^{-1}) = 1$, so every unit has norm 1. Similarly, if $N(u) = 1$, then $a = \pm 1, b = 0$, so every norm 1 element is a unit. Thus the only units in R are ± 1.

   Now suppose $2 = \alpha\beta$. Then $4 = N(\alpha)N(\beta)$, so $N(\alpha) = N(\beta) = 2$. However, for $n > 3$, this has no solutions, so one of them must have norm 1, implying it's a unit. Thus 2 is irreducible.

   Suppose $\sqrt{-n} = \alpha\beta$. Then $n = N(\alpha)N(\beta)$. If neither are 1, then $N(\alpha), N(\beta) < n$. Thus the $\sqrt{-n}$ component in each one must be zero, implying that both α and β are integers, which is clearly false. Thus one of α, β is a unit, so $\sqrt{-n}$ is irreducible. Similar logic shows that $1 + \sqrt{-n}$ and its conjugate are irreducible.

b) If n is even, then $-2 \cdot \frac{n}{2} = \left(\sqrt{-n}\right)^2 = -n$. and since 2 doesn't divide $\sqrt{-n}$, it's not a prime element. Since in a UFD every irreducible is prime, we see that $\mathbb{Z}[\sqrt{-n}]$ is not a UFD. If n is odd, then $2 \cdot \frac{n+1}{2} = (1 + \sqrt{-n})(1 - \sqrt{-n}) = n + 1$. Again 2 doesn't divide $1 \pm \sqrt{-n}$, so it's not a prime element. Thus $\mathbb{Z}[\sqrt{-n}]$ is not a UFD.

c) For even n, consider $I = (2, \sqrt{-n})$. Suppose for the sake of contradiction that $(2, \sqrt{-n}) = (\alpha)$. Thus $2 = \alpha\beta$, implying $4 = N(\alpha)N(\beta)$. From above, we see that this implies either α or β is a unit, so $(\alpha) = R$. Thus $2x + \sqrt{-n}y = 1$ for some $x, y \in R$. Multiplying both sides by $\sqrt{-n}$ yields $2\sqrt{-n}x - ny = \sqrt{-n}$. Since the left side is divisible by 2 while the right side isn't, we have a contradiction, so the ideal can't be principal. Similar logic shows that $J = (2, 1 + \sqrt{-n})$ is not principal.

3. Fields and Galois Theory

3.1. Field Extension Basics

Definition (characteristic): The *characteristic* of a field F, denoted $\mathrm{ch}(F)$, is the smallest positive integer n such that $\underbrace{1_F + \cdots + 1_F}_{n \text{ times}} = 0$. If no such integer exists, then the characteristic is 0.

For an integer n, let $n \cdot \alpha = \underbrace{\alpha + \cdots + \alpha}_{n \text{ times}}$ in a field F with $\alpha \in F$ (with $(-1) \cdot \alpha = -\alpha$).

Proposition (characteristic is prime): The characteristic of a field is either 0 or prime. Furthermore, if $\mathrm{ch}(F) = p$, then for all $\alpha \in F$, $p \cdot \alpha = 0$.

Proof: Suppose $\mathrm{ch}(F)$ is $n = ab$ for $a, b > 1$. Then $(a \cdot 1_F)(b \cdot 1_F) = ab \cdot 1_F = 0$. However, the two terms on the left are nonzero, which contradicts F being an integral domain (since it's a field). Thus n must be prime. The second statement follows from $p \cdot \alpha = p \cdot (\alpha \cdot 1_F) = (p \cdot 1_F) \cdot \alpha = 0$. ■

Definition (prime subfield): The *prime subfield* of a field F is the subfield of F generated by 1_F.

From the ring homomorphism $\varphi : \mathbb{Z} \to F$ given by $\varphi(n) = n \cdot 1_F$, by the first isomorphism theorem we have $\mathbb{Z}/\ker\varphi \cong \varphi(\mathbb{Z})$. If $\mathrm{ch}(F) = 0$, then $\ker\varphi = \{0\}$, and if $\mathrm{ch}(F) = p$, then $\ker\varphi = p\mathbb{Z}$. If the first case, since F is a field, it contains the element $\frac{a \cdot 1_F}{b \cdot 1_F}$ for nonzero b. Thus F contains an isomorphic copy of $\mathbb{Q}$. Similarly, if $\mathrm{ch}(F) = p$, then F contains an isomorphic copy of $\mathbb{F}_p$. Thus the prime subfield is either $\mathbb{Q}$ or $\mathbb{F}_p$.

Definition (field extension): If K is a field containing the subfield F, then K is said to be a *field extension* of F and is denoted K/F.

Note that we can treat K as a vector space over F (for example, $\mathbb{R}$ can be treated as a vector space over $\mathbb{Q}$).

Definition (degree of field extension): The *degree* of a field extension K/F, denote $[K : F]$, is the dimension of K as a vector space over F. The extension is said to be *finite* if $[K : F]$ is finite and *infinite* otherwise.

Proposition: Let F be a field and let $p(x) \in F[x]$ be an irreducible polynomial. Then there exists a field K containing an isomorphic copy of F in which $p(x)$ has a root.

Proof: Consider $K = F[x]/(p(x))$. Note that since $p(x)$ is irreducible, $(p(x))$ is a prime ideal. Since $F[x]$ is a PID, this means $(p(x))$ is maximal. Thus K is a field. We have the canonical projection $\pi : F[x] \to F[x]/(p(x))$ that's a homomorphism. If we restrict to $\pi|_F$, then we have a homomorphism of fields. Since the only ideals of F are 0 and F, this implies that $\pi|_F$ is either trivial or injective, and we know it can't be trivial since $\pi|_F(1) = \pi(1) = 1 + (p(x)) \neq 0$. Thus $\pi|_F$ is injective, and thus there exists an isomorphic copy of F in K.

To check that $p(x)$ has a root in K, let $X = \pi(x) = x + (p(x)) \in K$. Since π is a homomorphism and p is a polynomial,

$$p(X) = p(\pi(x)) = \pi(p(x)) = p(x) + (p(x)) = 0,$$

so X is a root of p in K. ■

Through the identification of the ismorphic copy of F, the result tells us that there is a field extension of F such that p has a root in it.

Proposition: Let $p(x) \in F[x]$ be an irreducible polynomial of degree n and let $K = F[x]/(p(x))$. Let $\theta \equiv x \pmod{p(x)}$ in K. Then the element $1, \theta, \theta^2, \ldots, \theta^{n-1}$ are a basis for K as a vector space over F, and $[K : F] = n$.

Proof: Let $a(x) \in F[x]$. Since $F[x]$ is a Euclidean domain, we can divide a by p, yielding

$$a(x) = q(x)p(x) + r(x)$$

such that $q, r \in F[x]$ and $\deg r < n$. Thus $a(x) \equiv r(x) \pmod{p(x)}$. Thus every residue class in $F[x]/(p(x))$ can be represented as a polynomial of degree less than n. Thus the images of $1, x, \ldots, x^{n-1}$, which are the powers of θ up to $n-1$, span K as a vector space over F.

To show they form a basis, suppose

$$b_0 + b_1\theta + \cdots + b_{n-1}\theta^{n-1} = 0$$

in K for $b_i \in F$ not all zero. Thus

$$b_0 + b_1 x + \cdots + b_{n-1}x^{n-1} \equiv 0 \pmod{p(x)}$$

in F, which implies $p(x)$ divides the polynomial in $F[x]$. However, this is impossible since the polynomial has nonzero degree less than n. Thus the powers of θ are linearly independent in K over F. Thus $[K : F] = n$, as desired. ■

The above result thus tells us that every element of $F[x]/(p(x))$ can be represented as a polynomial of degree less than n in $\theta \in K$, where $p(\theta) = 0$. Addition and multiplication work as expected, with both being reduced mod $p(x)$. Since this is field, we can also divide by as well.

Definition (generated field extension): Let K be a field extension of F, and let $\alpha, \beta, \ldots \in K$. The smallest subfield of K containing both F and $\alpha, \beta, \ldots$, denoted $F(\alpha, \beta, \ldots)$, is called the *field generated by* $\alpha, \beta, \ldots$ *over* F.

Definition (simple/primitive element): If the field K is generated by a single element α over F, i.e. $K = F(\alpha)$, then K is said to be a *simple* extension of F, and the element α is called a *primitive element.*

Proposition: Let F be a field and let $p(x) \in F[x]$ be an irreducible polynomial. Suppose K is an extension field of F containing a root α of $p(x)$ (which exists by the few previous results), Let $F(\alpha)$ denote the subfield of K generated over F by α. Then

$$F(\alpha) \cong F[x]/(p(x)).$$

Proof: Define $\varphi : F[x] \to F(\alpha)$ as $\varphi(q(x)) = q(\alpha)$. Note that $(p(x))$ is in the kernel of φ. Thus we obtain a homomorphism $\psi : F[x]/(p(x)) \to F(\alpha)$ by $\psi(q(x) + (p(x))) = \varphi(q(\alpha))$ (which is well-defined since every representative of $q(x) + (p(x))$ has the same value at α since $p(\alpha) = 0$). Since $F[x]/(p(x))$ is a field and this map is nonzero (it maps F to F), ψ must be injective. Since the image is a subfield of F containing both F and α, by definition of $F(\alpha)$, the image must be $F(\alpha)$. Thus ψ is an isomorphism. ■

Combining this with the previous proposition, we have the following:

Corollary: Assume the hypothesis of the previous proposition and assume p has degree n. Then

$$F(\alpha) = \{a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1} : a_0, a_1 \cdots, a_{n-1} \in F\} \subseteq K.$$

Example: Consider $F = \mathbb{Q}$ and $p(x) = x^3 - 2$, which is irreducible. Then $\mathbb{Q}[x]/(x^3 - 2) \cong \mathbb{Q}\left(\sqrt[3]{2}\right)$. However, if we let ω be a primitive cube root of unity, then $p\left(\omega\sqrt[3]{2}\right) = 0$, so $\mathbb{Q}[x]/(x^3 - 2) \cong \mathbb{Q}\left(\omega\sqrt[3]{2}\right)$. Thus $\mathbb{Q}\left(\sqrt[3]{2}\right) \cong \mathbb{Q}\left(\omega\sqrt[3]{2}\right)$.

The example above shows us that the roots of an irreducible polynomial are algebraically indistinguishable, in the sense that the fields obtain by adjoining a root of it produces isomorphic fields. In fact, we can make this precise.

Proposition: Let $\varphi : F \to F'$ be an isomorphism of fields. Let $p(x) \in F[x]$ be an irreducible polynomial and let $p'(x) \in F'[x]$ be the irreducible polynomial obtained by applying the map φ to the coefficients of $p(x)$. Let α be a root of $p(x)$ in some extension of F, and let β be a root of $p'(x)$ in some extension of F'. Then there's an isomorphism $\sigma : F(\alpha) \to F'(\beta)$ that takes α to β and such that $\sigma|_F = \varphi$.

Proof: φ induces a natural homomorphism from $F[x]$ to $F'[x]$ by applying it to the coefficients of elements in $F[x]$, and so this map takes the maximal ideal $(p(x))$ to the maximal ideal $(p'(x))$. Taking quotients yields a field isomorphism from $F[x]/(p(x))$ to $F'[x]/(p'(x))$. By the previous proposition, the first field is isomorphic to $F(\alpha)$ and the second field is isomorphic to $F'(\beta)$. Composing isomorphisms yields σ, which clearly restricts to φ on F. ■

Example: As an explicit example, let $\mathbb{Q} = F = F'$ and $p(x) = p'(x) = x^3 - 2$. Let $\alpha = \sqrt[3]{2}$ and $\beta = \omega\sqrt[3]{2}$, where ω is primitive third root of unity. Then by the above, $\mathbb{Q}(\alpha) \cong \mathbb{Q}(\beta)$.

3.2. Algebraic Extensions

Definition (algebraic/transcendental): Let K be an extension of a field F. An element $\alpha \in K$ is said to be *algebraic* over F is α is the root of some nonzero polynomial $f(x) \in F[x]$. If α is not algebraic over F, then α is said to be *transcendental* over F. The extension K/F is said to be *algebraic* if every element of K is algebraic over F.

Proposition (minimal polynomial): Let α be algebraic over F. Then there exist a unique monic polynomial $m_{\alpha,F}(x) \in F[x]$ which has α has a root. Furthermore, a polynomial $f(x) \in F[x]$ has α as a root if and only if $m_{\alpha,F}(x)$ divides $f(x)$ in $F[x]$.

Proof: Let $g(x) \in F[x]$ be a polynomial of minimal degree with root α (which by the definition of being algebraic must exist). Multiply by a constant such that g is monic. If were reducible, say $g(x) = a(x)b(x)$ with $a, b \in F[x]$ with degree less than g, then $0 = g(\alpha) = a(\alpha)b(\alpha)$, which implies one of a, b has α as a zero, which contradicts the minimality of the degree of α. Thus g is irreducible. Furthermore, if there existed some other polynomial monic irreducible $g' \in F[x]$ with same degree as g and α as a root, then $g(x) - g'(x)$ would have degree strictly smaller than $\deg g$ and have α as a root, contradiction. Thus g is the unique monic irreducible polynomial with α as a root.

Now suppose $f(x) \in F[x]$ has α as a root. By the Euclidean algorithm, we can write $f(x) = q(x)g(x) + r(x)$ with $\deg r < \deg g$. Then $0 = f(\alpha) = q(\alpha)g(\alpha) + r(\alpha) = r(\alpha)$, which is a contradiction unless $r = 0$. Thus $g \mid f$, as desired. The other direction follows easily. ■

Corollary: If L/F is an extension of fields and α is algebraic over both F and L, then $m_{\alpha,L}(x)$ divides $m_{\alpha,F}(x)$ in $L[x]$.

Proof: Follows since $m_{\alpha,F}$ is a polynomial in $L[x]$ with root α. ■

Definition (minimial polynomial): The polynomial $m_{\alpha,F}(x)$ (or just m_α if F is clear from context) is called the *minimal polynomial* for α over F. The *degree* of α over F is $\deg m_{\alpha,F}$.

Proposition (degrees match): Let α be algebraic over the field F and let $F(\alpha)$ be the field generated by α over F. Then

$$F(\alpha) \cong F[x]/(m_\alpha(x))$$

and

$$[F(\alpha) : F] = \deg m_\alpha = \deg \alpha.$$

Proof: Follows from previous results. ■

Example: By Eisenstein, the polynomial $x^n - 2$ for $n > 1$ in $\mathbb{Q}$ is irreducible. Thus

$$\left[\mathbb{Q}\left(\sqrt[n]{2}\right) : \mathbb{Q}\right] = n.$$

However, over $\mathbb{R}$, the minimal polynomial of $\sqrt[n]{2}$ is just $x - \sqrt[n]{2}$.

Proposition: The element α is algebraic over F if and only if the simple extension $F(\alpha)/F$ is finite.

Proof: If α is algebraic, then $[F(\alpha) : F] = \deg m_{\alpha,F}$ which is finite. Now suppose $F(\alpha)/F$ is finite with degree n. Then the elements $1, \alpha, ..., \alpha^n$ are linearly dependent over F, so

$$b_0 + b_1\alpha + \cdots + b_n\alpha^n = 0$$

for some set of not all zero $b_i \in F$. Thus there's a polynomial in $F[x]$ that has α as a root, so α is algebraic. ■

Corollary: If the extenion K/F is finite, then it is algebraic.

Proof: If $\alpha \in K$, then the subfield $F(\alpha)$ is a subspace of the vector space K over F. Thus $[F(\alpha) : F] \leq [K : F]$, so by the previous proposition, α is algebraic over F. ■

Proposition (product of degrees): Let $F \subseteq K \subseteq L$ be fields. Then

$$[L : F] = [L : K][K : L],$$

where the degrees can possibly be infinite.

Proof: First suppose $[L : K] = m$ and $[K : F] = n$ are finite. Let $\alpha_1, ..., \alpha_m$ be a basis for L over K and $\beta_1, ..., \beta_n$ a basis for K over F. Thus every element of L can be written as $a_1\alpha_1 + \cdots + a_m\alpha_m$, where $a_i \in K$. Since every element of K can be written as $b_1\beta_1 + \cdots + b_n\beta_n$ with $b_i \in F$, we can sub in these expressions for the $a_i \in K$ and obtain a linear combination of $\alpha_i\beta_j$ with coefficients in F. Thus $\alpha_i\beta_j$ for $1 \leq i \leq m, 1 \leq j \leq n$ span L as a vector space over F.

Now we show this list is linearly independent. Suppose

$$\sum_{i=1}^{m}\sum_{j=1}^{n} b_{i,j}\alpha_i\beta_j = 0$$

for $b_{i,j} \in F$. Letting $a_i := \sum_{j=1}^{n} b_{i,j}\beta_j$ yields elements in K and

$$\sum_{i=1}^{m} a_i\alpha_i = 0.$$

Since α_i is a basis for L over K, this implies that each $a_i = 0$. Thus

$$b_{i,1}\beta_1 + \cdots + b_{i,n}\beta_n = 0$$

for all $1 \leq i \leq m$. Since β_j is a basis for K over F, this implies $\beta_{i,j} = 0$ for all $1 \leq i \leq m, 1 \leq j \leq n$. Thus $\alpha_i\beta_j$ is a linearly dependent list, and thus a basis for L over F. Thus $[L : F] = mn$, as desired.

Now suppose $[K : F]$ is infinite. Thus there infinitely many elements of K that are linearly dependent over F, and thus infinitely many elements of L that are linearly dependent over F. Thus $[L : F]$ is infinite. Similarly, if $[L : K]$ is infinite, then there are infinitely many elements of L that linearly dependent over K, and thus infinitely many elements of L are linearly dependent over F, so $[L : F] = 0$. Finally, if $[L : F]$ is finite, then one of $[L : K], [K : F]$ must be infinite, since otherwise the first part of the proof shows that $[L : F]$ is finite. ■

Example: The element $\sqrt{2}$ is not contained in $\mathbb{Q}\left(\sqrt[3]{2}\right)$, as $\left[\mathbb{Q}\left(\sqrt{2}\right) : \mathbb{Q}\right] = 2$ does not divide $\left[\mathbb{Q}\left(\sqrt[3]{2}\right) : \mathbb{Q}\right] = 3$ (which by the previous result implies that $\mathbb{Q} \subseteq \mathbb{Q}\left(\sqrt{2}\right) \subseteq \mathbb{Q}\left(\sqrt[3]{2}\right)$ is impossible).

Example: We know the minimal polynomial of $\sqrt[6]{2}$ over $\mathbb{Q}$ is $x^6 - 2$, and clearly $\mathbb{Q}\left(\sqrt{2}\right) \subseteq \mathbb{Q}\left(\sqrt[6]{2}\right)$ (as $\left(\sqrt[6]{2}\right)^3 = \sqrt{2}$). Thus by the proposition,

$$\left[\mathbb{Q}\left(\sqrt[6]{2}\right) : \mathbb{Q}\left(\sqrt{2}\right)\right] = \left[\mathbb{Q}\left(\sqrt[6]{2}\right) : \mathbb{Q}\right] / \left[\mathbb{Q}\left(\sqrt{2}\right) : \mathbb{Q}\right] = 3.$$

Indeed, the polynomial $x^3 - \sqrt{2}$, which has root $\sqrt[6]{2}$, is irreducible in $\mathbb{Q}\left(\sqrt{2}\right)$ by Eisenstein with $\sqrt{2}$.

Definition (finitely generated extension): An extension K/F is *finitely generated* if there are elements $\alpha_1, \ldots, \alpha_k$ in K such that $K = F(\alpha_1, \ldots, \alpha_k)$.

Proposition: The generated over F by α and β is the field generated by β over $F(\alpha)$. In other words,

$$F(\alpha, \beta) = (F(\alpha))(\beta).$$

Proof: Clearly $(F(\alpha))(\beta)$ contains both α and β, so since $F(\alpha, \beta)$ by definition is the smallest field extension that contains both, we have $F(\alpha, \beta) \subseteq (F(\alpha))(\beta)$. Meanwhile, $(F(\alpha))(\beta)$ is the smallest field that contains $F(\alpha)$ and β, which $F(\alpha, \beta)$ clearly also does, so $(F(\alpha))(\beta) \subseteq F(\alpha, \beta)$. ■

Proposition: The extension K/F is finite if and only if K is generated by a finite number of algebraic elements over F.

Proof: Suppose $[K : F] = n$. Then there exists $\alpha_1, ..., \alpha_n \in K$ that are a basis for K over F. Then since $F \subseteq F(\alpha_i) \subseteq K$, we see that $[F(\alpha_i) : F] \leq n$. Since this simple extension is finite, it is algebraic, so α_i is indeed algebraic over F.

For the converse, we induct on the number of generating elements. In the base case, there's some generating algebraic element α. We know from previous results that $F(\alpha)/F$ is finite and has degree equal to the degree of α. Now suppose $K = F(\alpha_1, ..., \alpha_n)$ for algebraic α_i. By the previous proposition, we can write $K = (F(\alpha_1, ..., \alpha_{n-1}))(\alpha_n)$. Since α_n is algebraic over F, it's also algebraic over the extension field $F(\alpha_1, ..., \alpha_{n-1})$. Thus $K/F(\alpha_1, ..., \alpha_{n-1}) = (F(\alpha_1, ..., \alpha_{n-1}))(\alpha_n)/F(\alpha_1, ..., \alpha_{n-1})$ is finite. Since by the induction hypotheses, the extension $F(\alpha_1, ..., \alpha_{n-1})/F$ is finite, we see that

$$[K : F] = [K : F(\alpha_1, ..., \alpha_{n-1})][F(\alpha_1, ..., \alpha_{n-1}) : F]$$

is also finite, as desired. ■

Proposition: Suppose $F \subseteq K \subseteq L$ are fields. If L/K and K/F are algebraic extensions, then L/F is an algebraic extension.

Proof: Suppose $\alpha \in L$. Then there exists $a_0, ..., a_n \in K$ such that α is a root of $a_n x^n + \cdots + a_0$. Now consider $F(\alpha, a_0, ..., a_n)$. Since each of the a_i is algebraic over F by hypothesis, by the previous proposition, $F(a_0, ..., a_n)/F$ is finite. By the polynomial, we see that $(F(a_0, ..., a_n))(\alpha) = F(\alpha, a_0, ..., a_n)$ has at most degree n over $F(a_0, ..., a_n)$ (since the minimal polynomial $m_{\alpha, F(a_0, ..., a_n)}$ must divide the minimlal polynomial $m_{\alpha, K}$, since K is an extension of $F(a_0, ..., a_n)$, and thus must also divide any polynomial in K that has α as a root). Thus

$$[F(\alpha, a_0, ..., a_n) : F] = [F(\alpha, a_0, ..., a_n) : F(a_0, ..., a_n)][F(a_0, ..., a_n) : F]$$

is finite, and thus $F(\alpha, a_0, ..., a_n)$ is algebraic over F. In particular, this implies α is algebraic over F. Since α was arbitrary, we're done. ■

Definition (composite field): Let K_1, K_2 be subfields of K. Then the *composite field* of K_1 and K_2, denoted K_1K_2, is the smallest subfield of K containing both K_1 and K_2.

This definition is easily extended to any finite number of subfields.

3.3. Splitting Fields and Algebraic Closures

Definition (splitting field): The extension field K of F is called a *splitting field* for the polynomial $f(x) \in F[x]$ if $f(x)$ factors completely into linear factors (or *splits completely*) in $K[x]$ and does not factor completely into linear factors over any proper subfield of K containing F.

Proposition (splitting field exists): For any field f, if $f(x) \in F[x]$, then there exists a field extension K of F which is a splitting field for $f(x)$.

Proof: We first show an extension where f splits completely exists by induction on $\deg f$. Afterwards, we just take the intersection of all such fields to obtain K.

If $n = 1$, then we can just take $E = F$ to be a field over which f splits. Now suppose $\deg f = n > 1$. If the irreducible factors of f all have degree 1, just take $E = F$. Otherwise, at least one factor, say $p(x)$, has degree ≥ 2. Thus there exists $E_1 \supseteq F$ that contains a root α of $p(x)$. Thus over E_1, $f(x)$ splits into $x - \alpha$ and another factor, say $f_1(x)$. This factor has degree $n - 1$, so by the induction hypothesis, there exists $E \supseteq E_1$ containing all roots of $f_1(x)$. Since clearly $\alpha \in E$, we see that f splits completely over E, as desired. ■

Proposition: Let $\varphi : F \to F'$ be a field isomorphism. Let $f(x) \in F[x]$, and let $f'(x) \in F'[x]$ be the corresponding polynomial obtained by applying φ to the coefficients of $f(x)$. Let E be a splitting field for $f(x)$ over F and E' be a splitting field for $f'(x)$ over F'. Then φ extends to an ismorphism $\sigma : E \to E'$ such that $\sigma|_F = \varphi$.

Proof: We again induct on $\deg f$. Note that under φ, we clearly have a correspondence between factors of f and f'.

If f has all roots in $F[x]$, then f splits completely in F, and thus so thus f' in F'. Thus $E = F, E' = F'$, and we can take $\sigma = \varphi$. Now suppose f has an irreducible factors $p(x)$ of degree ≥ 2. Let p' be the corresponding irreducible factor of f'. If $\alpha \in E$ is a root of $p(x)$ and $\beta \in E'$ is a root of $p'(x)$, then by a previous result, there exists an ismorphism $\sigma' : F(\alpha) \to F'(\beta)$ that extends φ. Over these fields, f and f' split into at least a linear factor and a degree $n - 1$ factor. Let these other factors be f_1 and f_1'. If L was a splitting field for f_1 over F_1, then it would contain all the roots of f, and thus would be a splitting field for f over F (since $F \subseteq F_1$), and so $L = E$ is a splitting field for f_1. Similarly, E' is a splitting field for f_1'. Then by the induction hypothesis, there exists an isomorphism $\sigma : E \to E'$ extending σ'. Since $\sigma|_{F_1} = \sigma'$ and $\sigma'|_F = \varphi$, we see that $\sigma|_F = \varphi$, as desired. ■

Corollary: The splitting field for any $f(x) \in F[x]$ over F is unique up to isomorphism.

Proof: Take φ to be the identity map from F to F in the previous proposition. ■

We now list some examples computing the splitting field.

Example: Consider $x^2 - 2$ over $\mathbb{Q}$. Since the polynomial is irreducible, the splitting field obviously can't be $\mathbb{Q}$. Since $\mathbb{Q}\left(\sqrt{2}\right)$ has $\pm\sqrt{2}$ and is a degree 2 extension, there is no proper subfield over which $x^2 - 2$ splits, so $\mathbb{Q}\left(\sqrt{2}\right)$ is the splitting field of $x^2 - 2$ over $\mathbb{Q}$.

Example: Consider $x^3 - 2$ over $\mathbb{Q}$. Note that $\mathbb{Q}\left(\sqrt[3]{2}\right)$ isn't the splitting field, since it only contains real numbers, while the polynomial has complex roots. Letting $\omega = \exp\left(\frac{2\pi i}{3}\right)$, this polynomial has roots $\sqrt[3]{2}, \omega\sqrt[3]{2}, \omega^2\sqrt[3]{2}$. Thus the splitting field K must also contain $\omega\sqrt[3]{2}/\sqrt[3]{2} = \omega$, which implies $\mathbb{Q}(\omega) \subseteq K$. On the other hand, any field containing $\sqrt[3]{2}$ and ω will contain all the roots. Thus the splitting field will be the smallest field containing both $\sqrt[3]{2}$ and ω, which means $K = \mathbb{Q}\left(\sqrt[3]{2}, \omega\right)$. Note that $\mathbb{Q} \subseteq \mathbb{Q}\left(\sqrt[3]{2}\right) \subseteq K$. The first extension has degree 3, and the second has degree 2, since $x^2 + x + 1$ has only complex roots. Thus $[K : \mathbb{Q}] = 6$, so the splitting field has degree 6.

Example: More generally, consider $x^p - 2$ over $\mathbb{Q}$ for prime p. Let ζ_p denote a pth root of unity (which has minimal polynomial $\frac{x^p-1}{x-1}$). This polynomial has roots $\zeta_p^k \sqrt[p]{2}$ for $0 \leq k \leq p-1$. Again, we see that any splitting field must contain both $\sqrt[p]{2}$ and ζ_p, and that any field containing those two must contain the splitting field. Thus the splitting field is $\mathbb{Q}\left(\sqrt[p]{2}, \zeta_p\right)$. This field contains both $\mathbb{Q}\left(\sqrt[p]{2}\right)$ and $\mathbb{Q}(\zeta_p)$, which with respect to $\mathbb{Q}$ have degree p and $p-1$ respectively. Since $\left[\mathbb{Q}\left(\sqrt[p]{2}, \zeta_p\right) : \mathbb{Q}\left(\sqrt[p]{2}\right)\right]$ is at most the degree of the minimal polynomial of $\sqrt[p]{2}$, which is $x^p - 2$, we see that $\left[\mathbb{Q}\left(\sqrt[p]{2}, \zeta_p\right) : \mathbb{Q}\right]$ has degree at most $p(p-1)$, and thus must have degree $p-1$.

Definition (algebraic closure): The field $\overline{F}$ is called an *algebraic closure* of F is $\overline{F}$ is algebraic over F and if every polynomial $f(x) \in F[x]$ splits completely over $\overline{F}$.

Definition (algebraically closed): A field K is said to be *algebraically closed* if every polynomial with coefficients in K has a root in K.

Note that from this definition, in an algebraically closed field, every polynomial with coefficients in the field splits completely. From this we see that the algebraic closure of an algebraically closed is itself.

Proposition: Let $\overline{F}$ be an algebraic closure of F. Then $\overline{F}$ is algebraically closed.

From the previous statement, this shows that $\overline{\overline{F}} = \overline{F}$.

Proof: Let $f(x) \in \overline{F}[x]$, and let α be a root of $f(x)$. Then $\overline{F}(\alpha)$ is an algebraic extension of $\overline{F}$. Since $\overline{F}$ is algebraic over F, we see that $\overline{F}(\alpha)$ is algebraic over F. Thus α is algebraic over F. This implies that there's some polynomial over F which has root α, which by the definition of algebraic closure implies that $\alpha \in \overline{F}$. Thus $\overline{F}$ is algebraically closed. ■

Proposition: For any field F there exists an algebraically closed field K closed containing F.

Proof: For any nonconstant $f \in F[x]$, then x_f denote an indeterminate. Consider the polynomial ring $F[\ldots, x_f, \ldots]$ generated by these indeterminates for all such $f \in F[x]$. Consider the ideal I generated by the polynomials $f(x_f)$. If this ideal is not proper, then 1 is an element of the ideal, so we have

$$g_1 f_1(x_{f_1}) + \cdots + g_n f(x_{f_n}) = 1$$

for some polynomials $g_i \in F[\ldots, x_f, \ldots]$. Let $x_{f_i} = x_i$, and let the remaining variables in the g_i be $x_{n+1}, \ldots, x_m$. Let F' be a finite extension containing a root α_i for each f_i (which we already know exists). Then setting $x_i = \alpha_i$ for $i = 1, \ldots, n$ and setting every other variable to 0 yields $0 = 1$, contradiction.

Thus I can't be proper, and so is contained in some maximal ideal $\mathcal{M}$ (this requires Zorn's lemma). Let $K_1 = F[\ldots, x_f, \ldots]/\mathcal{M}$ be a field. Since $\mathcal{M}$ is proper, K_1 contains an isomorphic copy of F. By construction, each polynomial f has a root in K_1 (the image of x_f in the field). Thus every polynomial in F has a root in K_1. We can do the same process to obtain a field K_2 in which every polynomial in K_1 has a root in K_2. Continuing in this fashion, we obtain a sequence of fields

$$F = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots,$$

where all polynomials in K_i have a root in K_{i+1}. Let

$$K = \bigcup_{i=0}^{\infty} K_i,$$

which still contains an isomorphic copy of F. Let $h(x) \in K[x]$. Then all the coefficients must lie in maximal field K_N, so $h(x) \in K_N[x]$, so by construction, this has a root in K_{N+1}, and thus a root in K. Thus K is algebraically closed, as desired. ■

Proposition: Let K be an algebraically closed field and let F be a subfield of K. Then the collection of elements $\overline{F}$ of K that are algebraic over F is an algebraic closure of F.

Proof: Every polynomial $f \in F[x]$ splits completely over K into factors $x - \alpha$. Thus α is algebraic over F, and so is an element of $\overline{F}$. Thus f splits completely in $\overline{F}$, so $\overline{F}$ is an algebraic closure of F. ■

Thus whenever we work with the roots of polynomials over some field, we can always assume we're working with respect to the larger algebraic closure. As an example, we have the following:

Proposition: Suppose $f(x) \in F[x]$ for some field F. Let $\alpha_1, \ldots, \alpha_n$ be the roots of f in $\overline{F}$. Then $F(\alpha_1, \ldots, \alpha_n)$ is the splitting field of f over F.

Proof: Clearly since the splitting field K must contain $\alpha_1, \ldots, \alpha_n$, it will contain $F(\alpha_1, \ldots, \alpha_n)$. Since $F(\alpha_1, \ldots, \alpha_n)$ contains each root, f splits over the field, so we have $K \subseteq F(\alpha_1, \ldots, \alpha_n)$. ■

The algebraic closure is also unique up to isomorphism (SHOW LATER).

3.4. Separable and Inseparable Extensions

Definition (separable): A polynomial over F is called *separable* if in the algebraic closure of F, all the roots of the polynomial are distinct. Otherwise the polynomial is *inseparable*.

Example: $x^2 - 2$ is separable over $\mathbb{Q}$, but $(x^2 - 2)^n$ isn't.

Proposition: A polynomial $f(x)$ has a multiple root α if and only if α is also the root of $f'(x)$.

Here the derivative of a polynomial is purely symbolic.

Proof: If α is a multiple root of f, say $f(x) = (x-\alpha)^n g(x)$, then $f'(x) = n(x-\alpha)^{n-1}g(x) + (x-\alpha)^n g'(x)$, so $f'(\alpha) = 0$. Conversely, if f and f' have α as a root, then $f(x) = (x-\alpha)h(x) \Rightarrow f'(x) = h(x) + (x-\alpha)h'(x) \Rightarrow h(\alpha) = 0 \Rightarrow f(x) = (x-\alpha)^2 h_1(x)$. ■

Equivalently, f has a multiple root if and only if the minimal polynomial divides both it and its derivative. Another equivalence: a polynomial has a muliple root if and only if its relatively prime to its derivative.

Corollary: Every irreducible polynomial over a field of characteristic 0 is separable.

Proof: Let F have characteristic 0, and suppose $p(x) \in F[x]$ is irreducible with degree n. Then $f'(x)$ has degree $n-1$. If $\gcd(f, f')$ is not 1, then some nonconstant polynomial in F divides f, which contradicts irreducibility. Thus f, f' are relatively prime, and thus f is separable. ■

Proposition (freshman's dream): Let F be a field of characteristic p. Then for any $a, b \in F$,

$$(a+b)^p = a^p + b^p \quad \text{and} \quad (ab)^p = a^p b^p.$$

Proof: The second equation is trivial. The first follows from the binomial theorem and noting that $p \mid \binom{p}{k}$ for $1 \le k \le p-1$. ■

Proposition: Suppose $\mathbb{F}$ is a finite field of characteristic p. Then every element of $\mathbb{F}$ is a p-th power.

Proof: Consider $\varphi : \mathbb{F} \to \mathbb{F}$ given by $\varphi(a) = a^p$. Suppose $a^p = \varphi(a) = \varphi(b) = b^p$. By the previous proposition, $0 = a^p - b^p = (a-b)^p \Rightarrow a - b = 0$ (this works over $p = 2$ as well since signs don't matter then). Thus φ is injective. Since $\mathbb{F}$ is finite, this implies φ is bijective. ■

Proposition: Every irreducible polynomial over a finite field $\mathbb{F}$ is separable.

Proof: Suppose $\mathbb{F}$ has characteristic p and $q(x) \in \mathbb{F}[x]$ is irreducible. Suppose for the sake of contradiction that q was inseparable. If q contained some monomial of power not divisible by p, then $\deg q' \neq 0$. If $\gcd(q, q')$ is not 1, then some nonconstant polynomial divides q, which contradicts irreducibility. Thus every monomial in q must be a power of p, so we can write $q(x) = r(x^p)$, where $r(x) = a_m x^m + \cdots + a_0 \in \mathbb{F}[x]$. By the previous proposition, we can write $a_i = b_i^p$. Then

$$q(x) = r(x^p) = (b_m x^m)^p + \cdots + b_0^p = (b_m x^m + \cdots + b_0)^p = r(x)^p,$$

where the third equality follows from freshman's dream. Thus q is reducible, contradiction. ■

3.5. Galois Theory Fundamentals

We denote the group of automorphisms of a field K by $\mathrm{Aut}(K)$ (which is a group under function composition). Furthermore, given some $\sigma \in \mathrm{Aut}(K)$, σ is said to *fix* a subset F of K if $\sigma(a) = a$ for $a \in F$.

Proposition: Every automorphism fixes the prime subfield of K.

Proof: If K is characteristic 0, then $\sigma(n) = n$ for $n \in \mathbb{Z}$, which then implies $\sigma(r) = r$ for $r \in \mathbb{Q}$. Similarly, if K is characteristic p, then $\sigma(n) = n$ for $n \in \mathbb{F}_p$. ■

Thus $\mathrm{Aut}(\mathbb{Q}), \mathrm{Aut}(\mathbb{F}_p) = \{1\}$, where 1 represents the identity map.

Definition (fixed field automorphisms): Let K/F be a field extension. Then $\mathrm{Aut}(K/F)$ denotes the collection of automorphisms of of K which fix F.

Again it's easy to see that $\mathrm{Aut}(K/F)$ is a group under composition. Furthermore, if F is the prime subfield of K, then by the above we see $\mathrm{Aut}(K) = \mathrm{Aut}(K/F)$.

Proposition (automorphism permutes roots): Let K/F be a field extension, and let $\alpha \in K$ be algebraic over F. Then for any $\sigma \in \mathrm{Aut}(K/F)$, $\sigma(\alpha)$ is a root of the minimal polynomial of α over F.

Proof: Let $m_\alpha(x) = x^n + \cdots + a_0$ be the minimal polynomial of α. Thus $\alpha^n + \cdots + a_0 = 0$. Applying σ to both sides yields

$$\begin{aligned} 0 &= \sigma(\alpha^n) + \sigma(a_{n-1}\alpha^{n-1}) + \cdots + \sigma(a_1\alpha) + \sigma(a_0) \\ &= \sigma(\alpha)^n + a_{n-1}\sigma(\alpha)^{n-1} + \cdots + a_1\sigma(\alpha) + a_0 \\ &= m_\alpha(\sigma(\alpha)). \end{aligned}$$

■

Example: Suppose $\tau \in \operatorname{Aut}\left(\mathbb{Q}\left(\sqrt{2}\right)/\mathbb{Q}\right)$. By the above, we see that $\tau\left(\sqrt{2}\right) = \pm\sqrt{2}$. Then since τ fixes $\mathbb{Q}$, we see that

$$\tau\left(a + b\sqrt{2}\right) = a \pm b\sqrt{2}.$$

Thus the only two automorphisms are the identity and "conjugation". From this we see that $\operatorname{Aut}\left(\mathbb{Q}\left(\sqrt{2}\right)/\mathbb{Q}\right) \cong \mathbb{Z}/2\mathbb{Z}$.

Example: We claim $\operatorname{Aut}\left(\mathbb{Q}\left(\sqrt[3]{2}\right)/\mathbb{Q}\right)$ is the trivial group. Indeed, we see that $\sigma\left(\sqrt[3]{2}\right)$ must be one $\sqrt[3]{2}, \zeta_3\sqrt[3]{2}, \zeta_3^2\sqrt[3]{2}$. However, the last two aren't real, while $\mathbb{Q}\left(\sqrt[3]{2}\right)$ is a real field, so we must have $\sigma\left(\sqrt[3]{2}\right) = \sqrt[3]{2}$. Thus $\sigma\left(a + b\sqrt[3]{2} + c\sqrt[3]{4}\right) = a + b\sqrt[3]{2} + c\sqrt[3]{4}$, so σ is the identity map.

3.6. Problems

Problem: Show that $p(x) = x^3 + 9x + 6$ is irreducible in $\mathbb{Q}[x]$, and let θ be a root of $p(x)$. Find the inverse of $1 + \theta$ in $\mathbb{Q}(\theta)$.

Proof: It's irreducible by Eisenstein's using 3. ■

Problem: Determine the splitting fields and degrees of the following polynomials over $\mathbb{Q}$:

a) $x^4 - 2$
b) $x^4 + 2$
c) $x^4 + x^2 + 1$
d) $x^6 - 4$

Solution:

a) We claim that the splitting field is $K = \mathbb{Q}\left(\sqrt[4]{2}, i\right)$. Clearly this contains all the roots. If some field contained $i\sqrt[4]{2}$ and $\sqrt[4]{2}$, then it contains i, so the splitting field contains $\mathbb{Q}\left(\sqrt[4]{2}, i\right)$, and thus actually is this field. Note that $[K : \mathbb{Q}] = \left[K : \mathbb{Q}\left(\sqrt[4]{2}\right)\right]\left[\mathbb{Q}\left(\sqrt[4]{2}\right) : \mathbb{Q}\right] = 2 \cdot 4 = 8$.

b)

Problem: Let K be a finite extension of F. Prove that K is a splitting field over F if and only if every irreducible polynomial in $F[x]$ that has a root in K completely splits over K.

Proof: First suppose K is a splitting field over F, say for some polynomial $f(x) = x^n + \cdots + a_0 \in F[x]$. Let $\alpha_1, \ldots, \alpha_n$ be the roots of f. Then $K = F(\alpha_1, \ldots, \alpha_n)$. Suppose $g(x) = x^m + \cdots + b_0 \in F[x]$ is irreducible with roots $\beta_1, \ldots, \beta_m$ and such that $\beta_1 \in K$. Note that $F(\beta, \alpha_1, \ldots, \alpha_n) = F(\alpha_1, \ldots, \alpha_n)$, since $\beta \in F(\alpha_1, \ldots, \alpha_n)$, so K is the splitting field of f over $F(\beta_1)$.

Pick $i = 2, \ldots, m$. Then since g is irreducible, there exists a field isomorphism $\varphi : F(\beta_1) \to F(\beta_i)$. Since these fields are isomorphic, the splitting field of $f(x)$ over each of these fields is also isomorphic, i.e. $\sigma : K \to K_i$ where σ is a field ismorphism and $\sigma|_{F(\beta_1)} = \varphi$. In particular, $K = F(\alpha_1, \ldots, \alpha_n) \subseteq K_i = F(\beta_i, \alpha_1, \ldots, \alpha_n)$. Since these fields are isomorphic, we must have

$[K_i : K] = 1$, which implies $\beta_i \in K$ (since otherwise the degree of K_i/K would be greater than 1). This works for all i, so g splits over K, as desired.

Now suppose that every irreducible polynomial in $F[x]$ that has a root in K splits over K. Let $G = \{g \in F[x] : g \text{ is irreducible and has a root in } K\}$. Let K_g denote the splitting field of g over F. Then by hypothesis, we see that $K_g \subseteq K$ for all $g \in G$.

Since K is finite, there is some basis $a_1, ..., a_n \in K$ for K/F. By finiteness, K is also algebraic, so there exists $g_1, ..., g_n \in F[x]$ for which g_i is the minimal polynomial of a_i over F. Let $h = g_1 \cdots g_n$. Let K_h be the splitting field of h. We claim that $K_h = K$, thus showing K is the splitting field of h over F.

Since $K_{g_i} \subseteq K$, we clearly have $K_{g_1} \cdots K_{g_n} \subseteq K$. Since every basis element a_i lies in some K_{g_i}, we also see that $K_{g_1} \cdots K_{g_n} \supseteq K$. Thus both fields are equal, so we just need to show that $K_h = K_{g_1} \cdots K_{g_n}$. Clearly $K_{g_1} \cdots K_{g_n} \subseteq K_h$. Let α be a root of h. Then it must be a root of one of the factors g_i, so $\alpha \in K_{g_i} \subseteq K_{g_1} \cdots K_{g_n}$. Thus $K_h \subseteq K_{g_1} \cdots K_{g_n}$, so we're done. ■

Remark: One of defintion of a normal extension is that an extension is normal if, when a polynomial has a root in the extension, the polynomial splits in the extension. The problem above shows that finite extensions are splitting fields if and only if they're normal.

Problem: Calculate $\mathrm{Aut}(\mathbb{R}/\mathbb{Q})$.

Solution: Since $\mathbb{Q}$ is a prime subfield of, every automorphism automatically fixed $\mathbb{Q}$. Let $\sigma \in \mathrm{Aut}(\mathbb{R}/\mathbb{Q})$. Note that $\sigma(\pm x^2) = \pm\sigma(x)^2$ for all $x \in \mathbb{R}$. Thus the sign of a real stays fixed. Suppose $0 < x < y$. Then $y - x > 0$, so $\sigma(y - x) > 0 \Rightarrow \sigma(y) > \sigma(x)$. Similarly we can show that $y < x < 0 \Rightarrow \sigma(x) > \sigma(y)$. Since we already saw that $x < 0 < y \Rightarrow \sigma(y) > 0 > \sigma(x)$, we see that σ is order preserving.